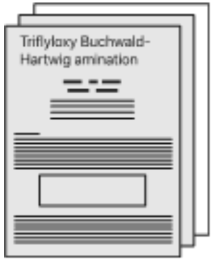
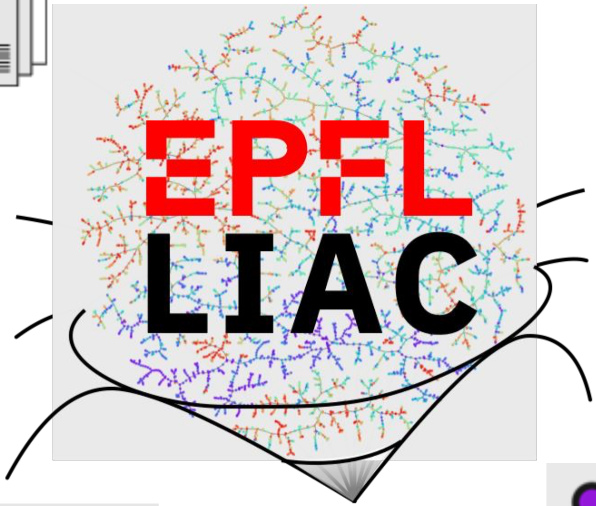
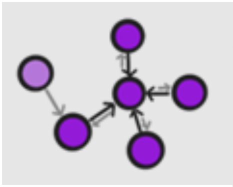
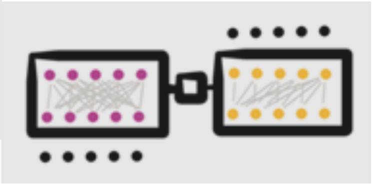


US20030166932A1: General Procedure H A solution of trifluoromethanesulfonic acid 3,5,8,8-tetramethyl-7,8-dihydronaphthalen-2-yl ester (Compound 35, 0.41 g, 1.2 mmol), Pd(OAc)2 (0.027 g, 0.12 mmol), BINAP (0.11 g, 0.18 mmol), Cs2CO3 (0.56 g, 1.72 mmol), ethyl 4-aminobenzoate (0.25 g, 1.5 mmol) and 5 mL of toluene was flushed with argon for 10 min, then stirred at 100° C. in a sealed tube for 48 h. ...







Language models for chemical sciences

Philippe Schwaller (he/him/his)

Laboratory of Artificial Chemical Intelligence (LIAC) – SB-ISIC

Background

- BSc/MSc in Materials Science (EPFL, '16)
- MPhil in Physics (Uni Cambridge, '19)
- PhD in Chemistry (Uni of Bern, '21)

IBM **Research**

- Researcher in ML for Chemistry ('17-'21)

Grew up in Switzerland
- French
- Swiss German / German

NCCR Catalysis

TT Assistant Professor
in Digital Chemistry
since Feb 2022

EPFL LIAC

Collaboration with synthetic chemists

Prix Schläfli since 1866

UNIVERSITÄT BERN

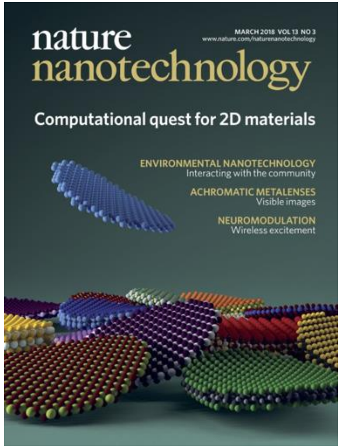
PhD in Chemistry
and Molecular Sciences ('21)
Prof Jean-Louis Reymond

EPFL

MANCHESTER 1824
The University of Manchester

MPhil in Physics ('19)
Dr Alpha Lee

UNIVERSITY OF CAMBRIDGE

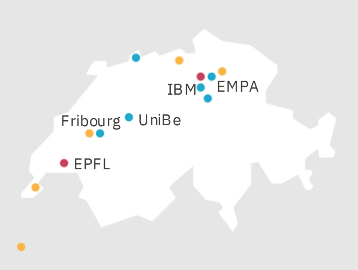


- MaX prize for frontier HPC applications ('17)
- PRACE HPC Excellence Award ('22)

Materials Science & Engineering

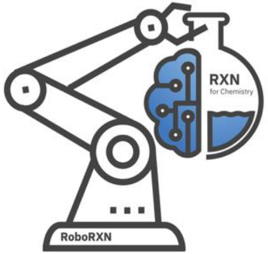
Virtual screening & simulation workflows
Prof Nicola Marzari

BSc ('14)/MSc ('16)



Empa
Materials Science and Technology

Lab work on ternary polymer
blends for organic solar cells
Prof Frank Nüesch



Machine learning for chemical synthesis
Intern/PhD/Postdoc
Dr Teodoro Laino

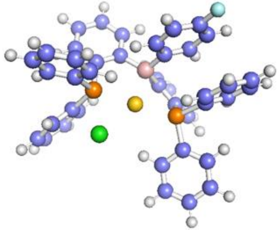
IBM **Research** Europe

Milky-way above the Swiss Alps, Schafarnisch, 2022

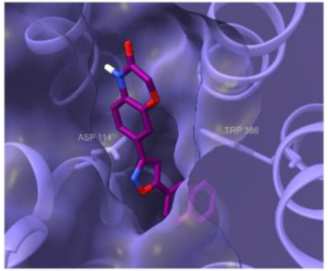
Chemical design cycle

What molecule to make?

How to make it?



Catalysis

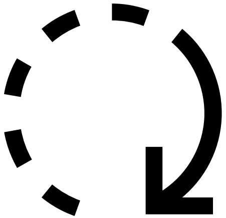


Drug discovery

Saturn (Sample-efficient de novo molecular design, with small language models)

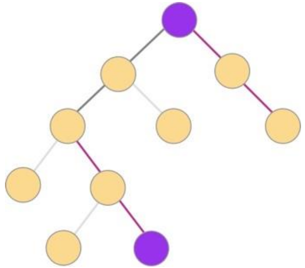
Design

Make



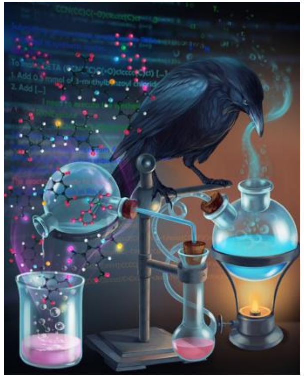
Test

Synthesis optimization



Synthesis planning

Experimental validation



ChemCrow (LLM agents for chemical research)

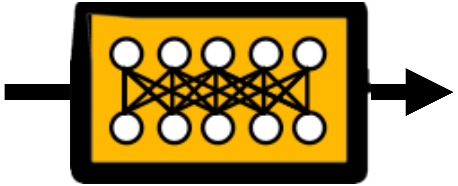
What do I mean by Language Model?

Context

Philippe

Philippe should

Philippe should prepare



Output

should

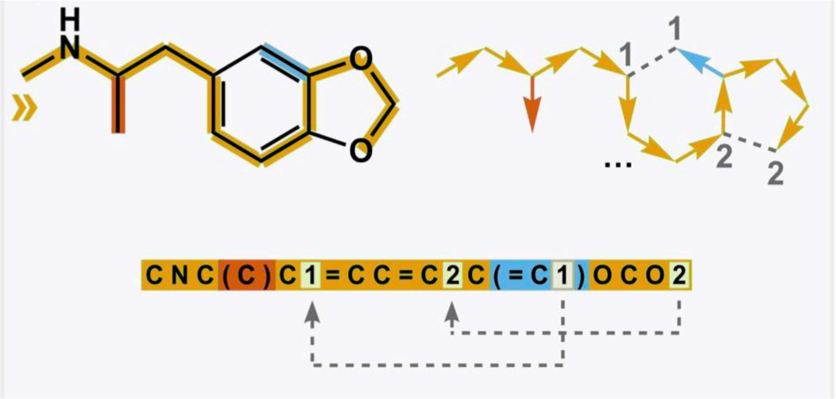
prepare

slides

… for the workshop in Zaragoza.

- **Trained to predict the next most likely word (in Chemistry, the next atom).**

Text-based molecule representation – SMILES



SMILES, a Chemical Language and Information System. 1. Introduction to Methodology and Encoding Rules

DAVID WEININGER

Medicinal Chemistry Project, Pomona College, Claremont, California 91711

Received June 17, 1987

Borrow methods for natural language and apply them to chemistry.

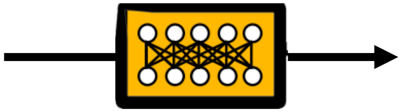
(e.g. Molecular Transformer, ACS Cent. Sci., Schwaller et al., 2019, first Transformer in Chemistry)

Teaching Language Models to Generate Molecules

PubChem

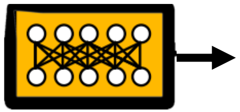
ChEMBL

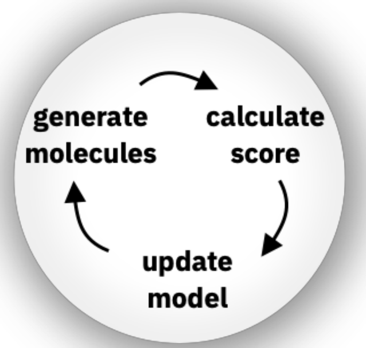
as SMILES



Model learns to **generate similar molecules – atom-by-atom**.

Designing molecules with target property profile (multi-objective)





Goal-directed learning

But scoring/labelling is expensive.

Augmented Memory & Saturn,
→ **Sample-efficient learning**
Jeff Guo

Tackling sample efficiency for high-fidelity feedback

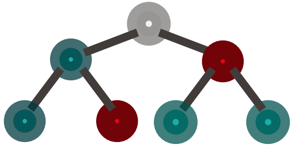
Oracle: Computational prediction or simulation

Sample Efficiency: How few oracle evaluations are required to optimize the objective?

Increasing predictive accuracy *but* also computational cost

Drug Discovery

Predictive Models



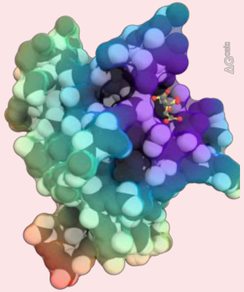
*Can be accurate but may have narrow domain of applicability

Molecular Docking

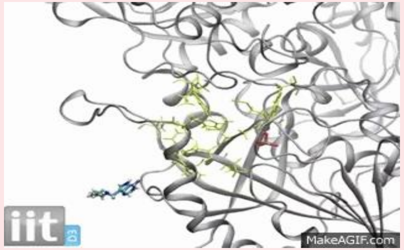
OpenEye SCIENTIFIC

Schrödinger

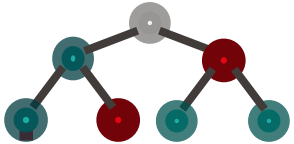
MMPB(GB)SA



Free Energy Calculations



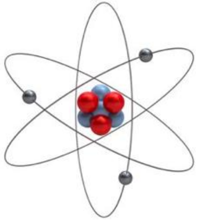
Materials and Catalyst Design



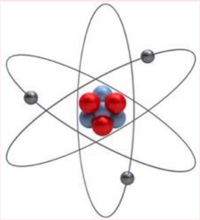
Semi-empirical QM

xTB

Single-point DFT



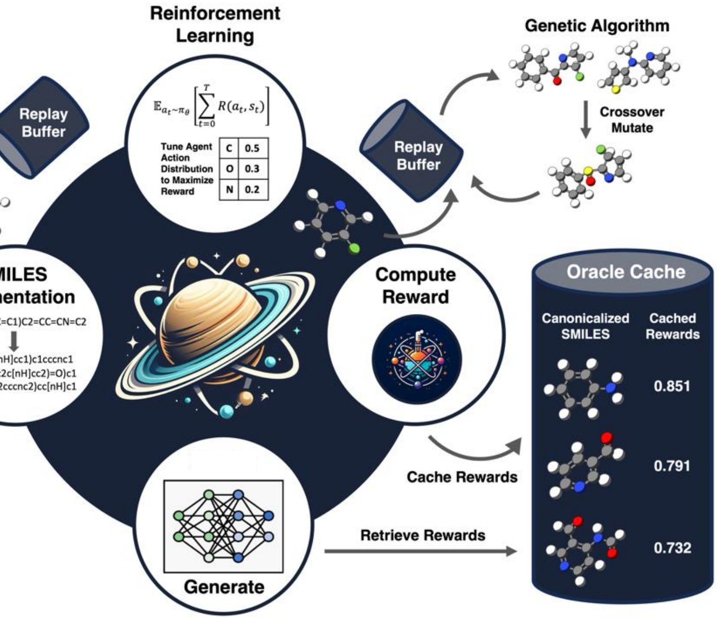
DFT – Varying Functionals / MD



Tackling sample efficiency for high-fidelity feedback

Augmented Memory (based on Reinvent, AZ)

• State of the art in sample efficiency (when published)



Guo, J. & Schwaller, P. *JACS Au*, **2024**.

Sample efficiency benchmark (PMO, NeurIPS `22)

Model	Rank (/30)	Score
Augmented Memory	1	15.002
REINVENT [4]	2	14.016
SynNet [8] [ICLR `22]	12	11.498
DoG-Gen [9] [NeurIPS `20]	13	11.456
DST [10] [ICLR `22]	15	10.989
MARS [11] [ICLR `21]	16	10.989
MIMOSA [12] [AAAI `21]	17	10.651
DoG-AE [9] [NeurIPS `20]	20	9.790
GFlowNet [13] [NeurIPS `21]	21	9.131
GA+D [14] [ICLR `20]	22	8.964
GFlowNet-AL [13] [NeurIPS `21]	27	8.406
JT-VAE [15] [ICML `18]	28	8.358

RNN-based

Approaches from big machine learning conferences.

10k oracle calls, 23 tasks
PMO by Coley group

Saturn – our Mamba-based de novo design framework

Multi-objective docking optimization (docking score, MW < 500 Da, maximize QED), reward threshold 0.8, **1000 oracle calls,** 10 seeds**.**

Target	Model	Yield (↑)	OB 1 (↓)
DRD2	Augmented Memory	22 ± 7	$143 \pm 75(10)$
	Saturn	369 ± 62	$93 \pm 53(10)$
AChE	Augmented Memory	173 ± 19	$57 \pm 2(10)$
	Saturn	480 ± 79	$32 \pm 24(10)$

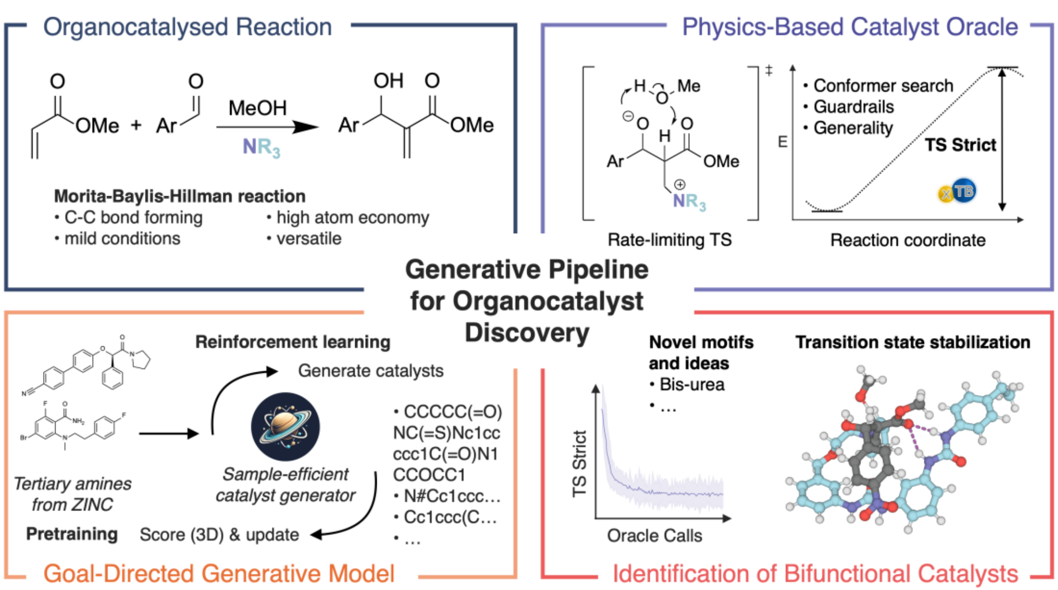
Saturn
- **<100 simulations** to find first candidate above the threshold.
- **>350 candidate molecules** with **1k simulations**

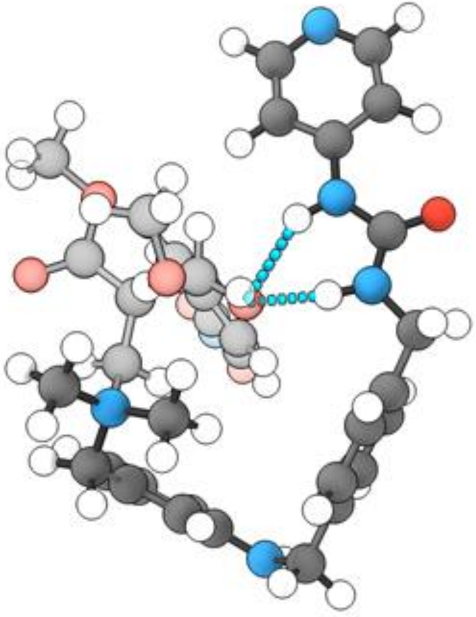
With Saturn (Mamba-based, state space model) compared to Augmented Memory (RNN-based)
- **Yield**, number of molecules above reward threshold, **increases**
- **Oracle Burden (OB)**, number to get to X molecules above threshold, **decreases**

Guo, J. & Schwaller, P. *ArXiv*, **2024**

https://github.com/schwallergroup/saturn

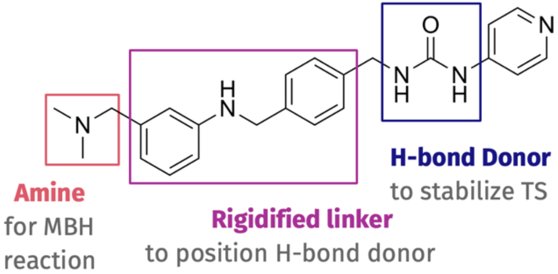
Saturn for (organo)catalysis





- Learns 3D functionality
- Language model as an idea generator
- WARNING: Your oracle will be exploited!
- Top candidates iteratively being synthesised (Nevado group, UZH)

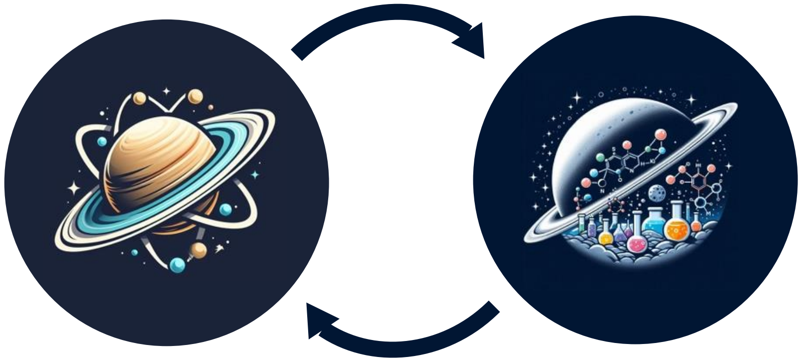
- Sarina Kopf



Key bottleneck for experimental validation → synthesizability…

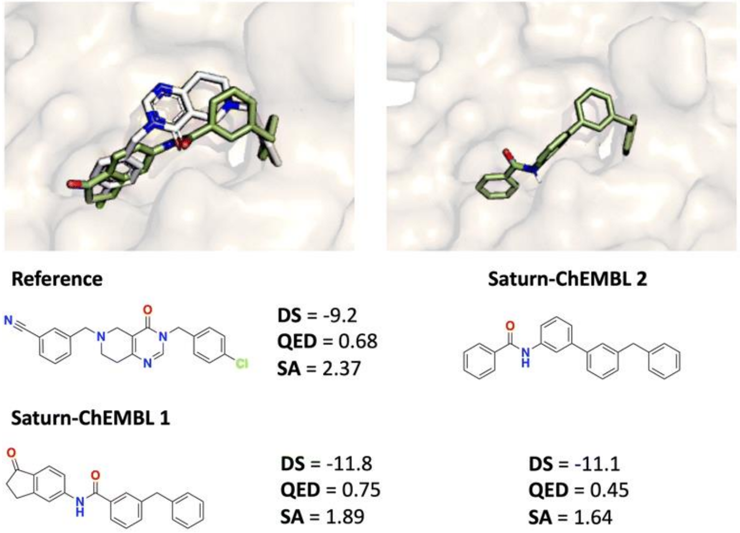
Sample efficiency bridges synthesizability

Why not run retrosynthesis on every generated molecule? The generative model must be ***sufficiently*** sample-efficient for this to be ***practically*** feasible



Saturn

Retrosynthesis model



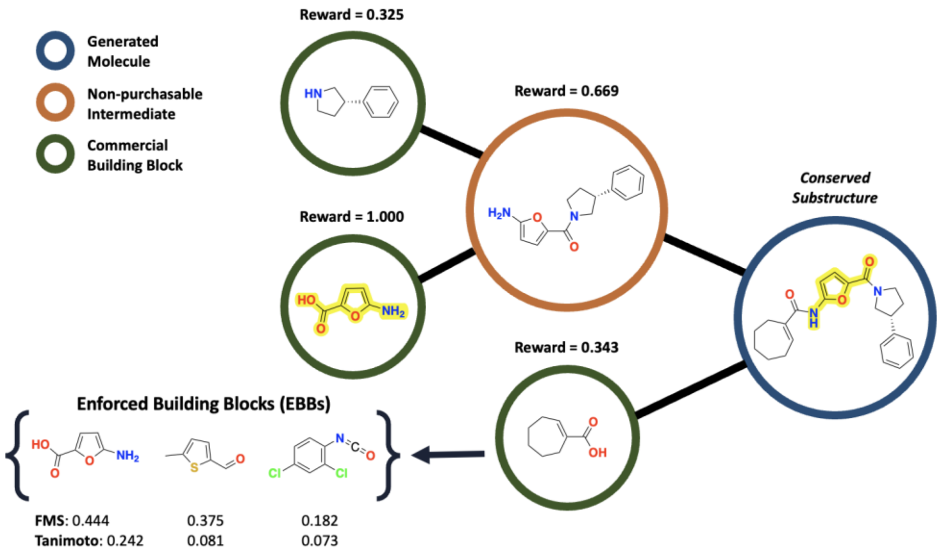
- 1000 oracle calls budget
- ClpP docking, QED, SA **+ valid route**

Optimizing for synthesizability: Guo & Schwaller *Chem Sci,* **2025**
TANGO: Guo & Schwaller *Nat. Comput. Sci. (accepted),* **2024**
Synthesizability control: Guo[*], Sabanza-Gil[*], et al. *arXiv,* **2025**

Constrained "synthesisability" – enforcing blocks

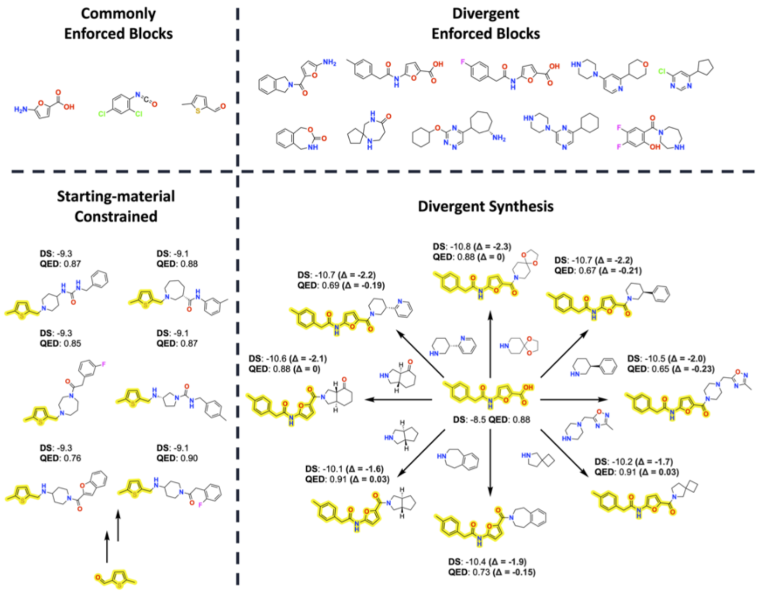
Challenge: Finding an **enforced building block in route** and rewarding that, makes **the reward landscape sparse**, and optimization is not feasible under a limited budget.

Solution: TANimoto Group Overlap (TANGO) reward → Tanimoto similarity + substructure match



TANGO Reward Function

$$\boldsymbol{TANGO}(\boldsymbol{x}, \boldsymbol{EBBs}) = \left(\max\left(\boldsymbol{FMS}(x, \boldsymbol{EBBs})\right) * Weight_{FMS}\right) + \left(\max\left(\boldsymbol{TanSim}(x, \boldsymbol{EBBs})\right) * Weight_{TanSim}\right) \in [0,1]$$



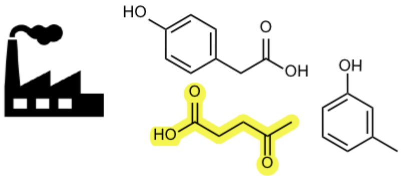
Multi-parameter optimization
(ClpP docking, QED, AiZynth with enforced BB)

Guo, J. & Schwaller, P. *ArXiv*, **2024**. (Accepted in Nature Computational Science)

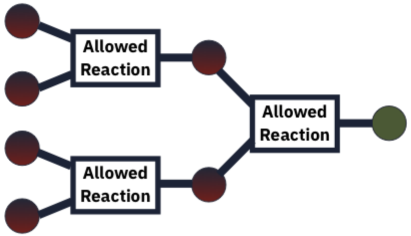
How to facilitate the synthesis even more?

Enforced building blocks

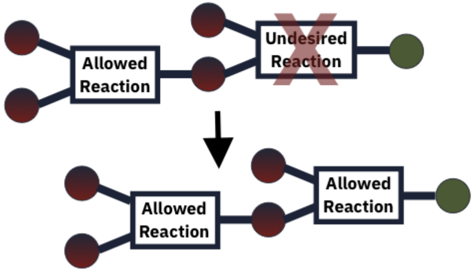
Additionally incorporate *specific* blocks in the synthesis *e.g.: industrial waste*



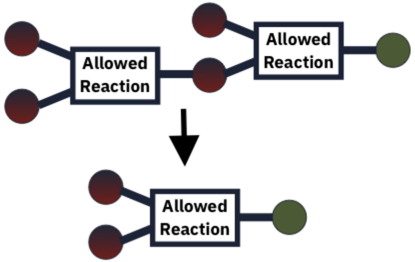
Enforce specific reactions



Avoid specific reactions



Minimize synthetic steps



Víctor Sabanza Gil

Jeff Guo

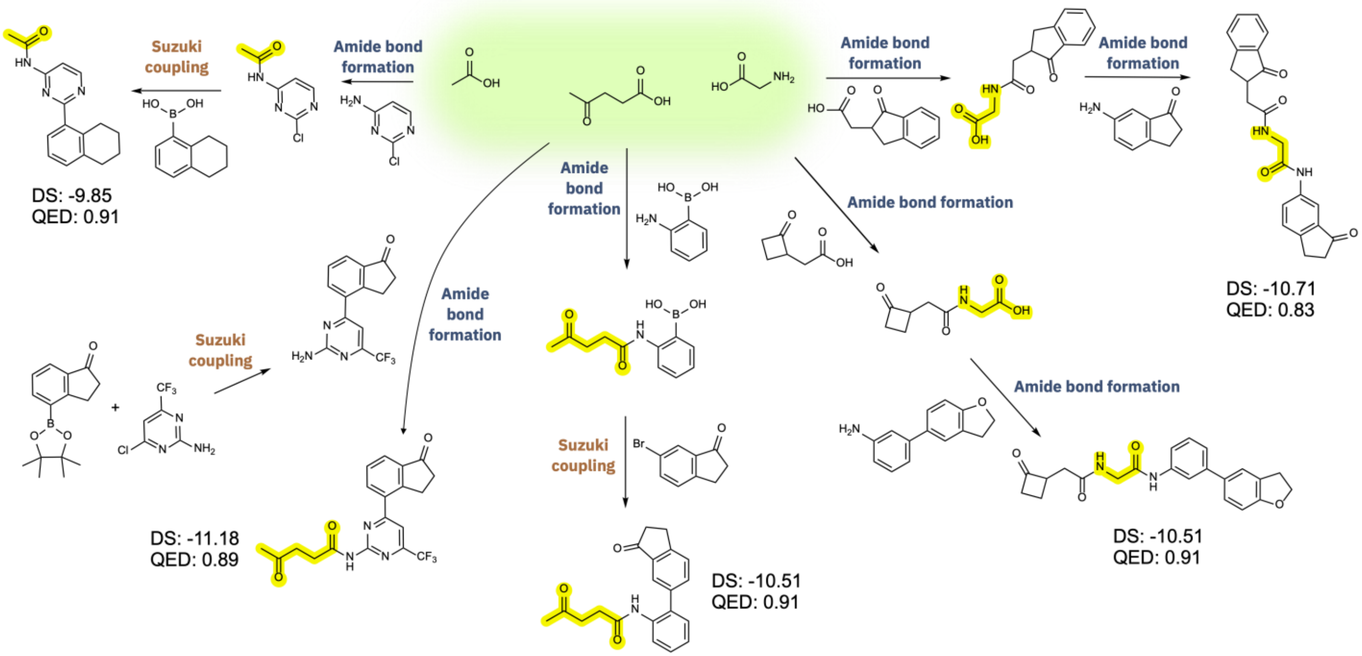
Steerable & granular synthesisability

Work by Jeff Guo and Victor Sabanza Gil.
https://arxiv.org/abs/2505.08774

Enforced set of building blocks (e.g., waste materials, bio-derived, or simply available in your lab)

EU STOCK

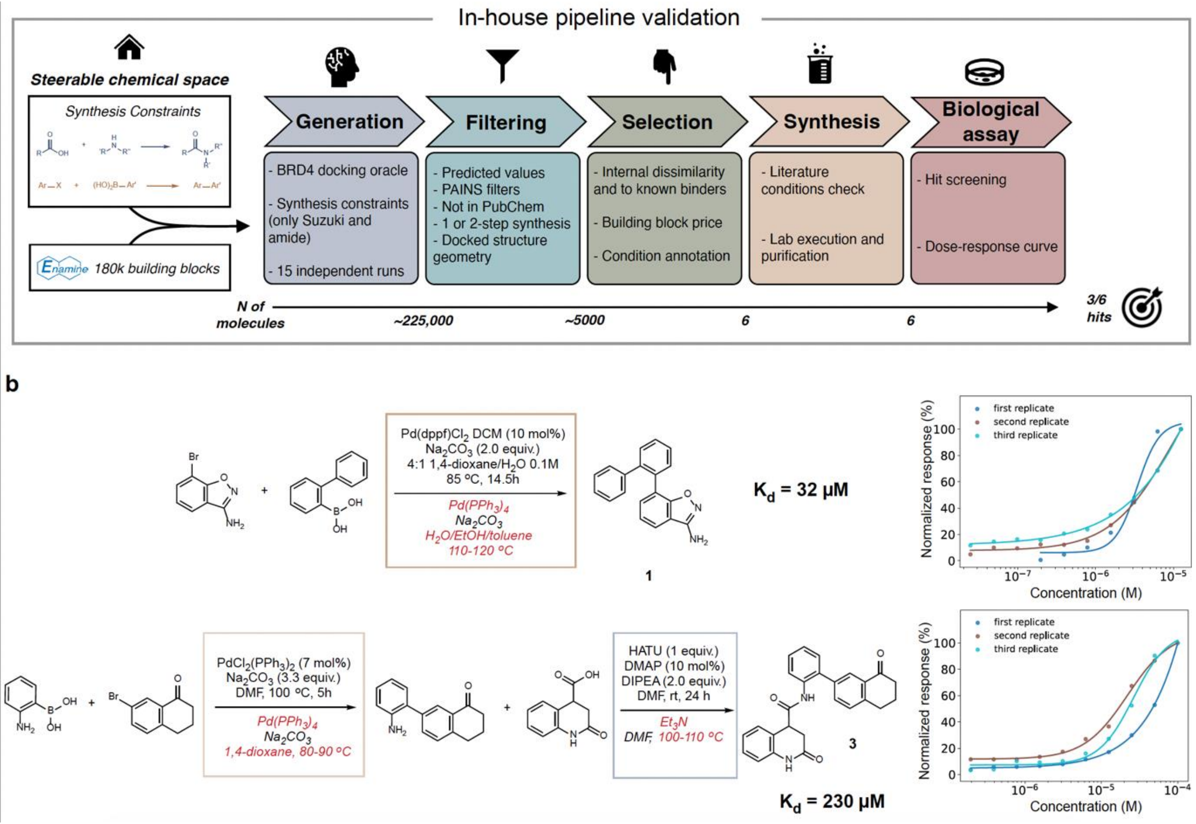
156k building blocks



Route constraints: Only **amide bond formations** and **Suzuki couplings** (your favourite reactions).
Objective: All constraints + low docking score (here against COX2) and high QED.
(Experimental validation in Waser and Luterbacher lab on-going)

EPFL

Experimental validation (BRD4) – with Waser/Luterbacher/Correia

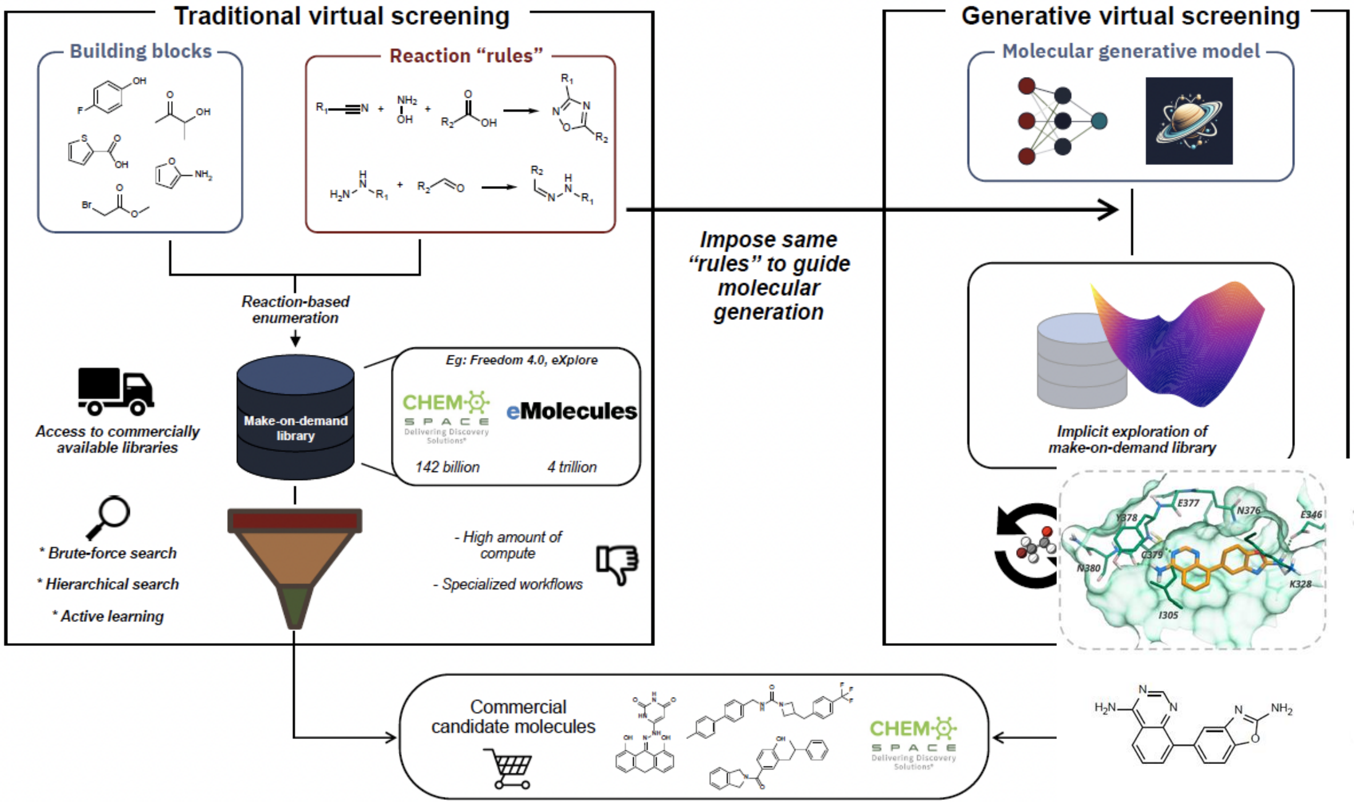


6 out of 6 syntheses worked.

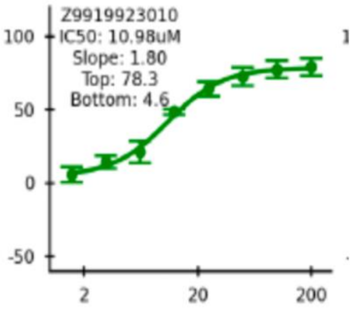
3/6 candidates showed at least some activity.

All open source oracles (better oracles would lead to better results)

Generative Virtual Screening (Wee1) – ChemSpace



- *Screened 225k/142B 0.00016% of the commercial library*
- *1 single consumer-grade GPU*



- **uM inhibitor found**

Sampling molecular space works.
How do we develop more realistic scorers?

nature machine intelligence

Review article

https://doi.org/10.1038/s42256-024-00843-5

Machine learning-aided generative molecular design

Received: 19 July 2023

Accepted: 24 April 2024

Published online: 18 June 2024

Check for updates

Yuanqi Du[1,10], Arian R. Jamasb[2,3,9,10], Jeff Guo[4,5,10], Tianfan Fu[6], Charles Harris[3], Yingheng Wang[1], Chenru Duan[7], Pietro Liò[3], Philippe Schwaller[4,5] & Tom L. Blundell[8]

Machine learning has provided a means to accelerate early-stage drug discovery by combining molecule generation and filtering steps in a single architecture that leverages the experience and design preferences of medicinal chemists. However, designing machine learning models that can achieve this on the fly to the satisfaction of medicinal chemists remains a challenge owing to the enormous search space. Researchers have addressed de novo design of molecules by decomposing the problem into a series of tasks determined by design criteria. Here we provide a comprehensive overview of the current state of the art in molecular design using machine learning models as well as important design decisions, such as the choice of molecular representations, generative methods and optimization strategies. Subsequently, we present a collection of practical applications in which the reviewed methodologies have been experimentally validated, encompassing both academic and industrial efforts. Finally, we draw attention to the theoretical, computational and empirical challenges in deploying generative machine learning and highlight future opportunities to better align such approaches to achieve realistic drug discovery end points.

Table 4 | Experimentally validated small-molecule generative design case studies

Model	Input	Output	Design task	Target	Hit rate	Outcome	Publication year
Distribution learning							
LSTM RNN[146]	SMILES	SMILES	De novo	RXR	4/5 (80%)	nM agonist	2018
LSTM RNN[147]	SMILES	SMILES	De novo	RXR	2/4 (50%)	µM agonist	2018
GraphGMVAE[148]	Graph	SMILES	Scaffold hopping	JAK1	7/7 (100%)	nM inhibitor	2021
LSTM RNN[108]	SMILES	SMILES	De novo	LXR	17/25 (68%)	µM agonist	2021
LSTM RNN[149]	SMILES	SMILES	De novo	RORγ	3/3 (100%)	µM agonist	2021
LSTM RNN[150]	SMILES	SMILES	De novo	FLT-3	1/1 (100%)	µM inhibitor	2022
GGNN GNN[151]	Graph	Graph	Fragment linking	CDK8	9/43 (21%)	nM inhibitor	2022
GRU RNN[152]	SMILES	SMILES	De novo	Bacteria	0/1 (0%)[a]	µM inhibitor	2022
BiRNN encoder–decoder[153]	SMILES	SMILES	De novo	DDR1	2/2 (100%)	nM inhibitor	2021
GRU RNN[154]	SMILES	SMILES	Reaction-based de novo	MERTK	15/17 (88%)	µM inhibitor	2022
LSTM RNN[155]	SMILES	SMILES	De novo	PI3Kγ	3/18 (17%)	nM inhibitor	2023
Transformer[156]	SMILES	SMILES	Fragment linking	TBK1	1/1 (100%)	nM inhibitor	2023
VAE and transformer[157]	SMILES	SMILES	Fragment hopping/linking	CDK2	17/23 (74%)[c]	nM inhibitor (MC)[b]	2023
LSTM RNN[102]	SMILES	SMILES	De novo	Nurr1γ	2/6 (33%)	nM inhibitor	2023
Graph transformer-LSTM RNN[158]	Graph	SMILES	De novo	PPARγ	2/2 (100%)	µM agonist	2023
Goal oriented							
DNC[159]	SMILES	SMILES	De novo	Kinases	0[d]	µM inhibitor	2018
AAE (conditional)[160]	SMILES	SMILES	De novo	JAK3	1/1 (100%)	µM inhibitor	2018
VAE[19]	SMILES	SMILES	De novo	DDR1	4/6 (67%)	nM inhibitor[a]	2019
LSTM RNN[108]	SMILES	SMILES	De novo ligand based	DDR1	4/6 (67%)	nM inhibitor	2021
Stack-GRU RNN[161]	SMILES	SMILES	De novo	EGFR	4/15 (27%)	nM inhibitor	2022
LSTM RNN (conditional)[107]	SMILES	SMILES	De novo	RIPK1	4/8 (50%)	nM inhibitor[a]	2022
Chemistry42[20]	Mixed	Mixed	De novo structure based	CDK20	6/13 (46%)[a]	nM inhibitor	2023
Chemistry42[162]	Mixed	Mixed	De novo structure based	CDK8	1/1 (100%)	nM inhibitor[a]	2023
Chemistry42[105]	Mixed	Mixed	De novo structure based (R-group)	SIK2	6/6 (100%)	nM inhibitor	2023
VAE[163]	SMILES	SMILES	De novo structure based	KOR	2/5 (40%)	µM antagonist	2023
Chemistry42[164]	Mixed	Mixed	De novo structure based	PHD enzymes	1/1 (100%)	nM inhibitor[a]	2024
GRU RNN-transformer[165]	SMILES	SMILES	De novo activity model	NLRP3	0[e]	nM inhibitor[a]	2024
Transformer-VAE (conditional)[166]	Geometry-SMILES	SMILES	De novo	Tuberculosis ClpP	1/6 (17%)[a]	µM inhibitor	2024
QC-LSTM RNN-Chemistry42[167]	SMILES	SMILES	De novo structure based	KRAS	1/12 (8%)[a]	µM inhibitor	2024
Graph transformer[168]	Graph	Graph	De novo activity model	MGLL	1/3 (33%)[a]	µM inhibitor	2024
Chemistry42[169]	Mixed	Mixed	Fragment linking	Polθ	4/6 (67%)	µM inhibitor[a]	2024
Chemistry42[14,15]	Mixed	Mixed	De novo structure based	TNIK	Unknown[f]	nM inhibitor[a]	2024
Attention-convolution layers[170]	Substructure vector	SMILES	Scaffold based	Factor Xa	Unknown[g]	µM inhibitor	2024
Flow (conditional)[171]	Geometry	Geometry	De novo	HAT1 and YTHDC1	0/2 and 0/3 (0%)[a]	Both µM inhibitor[a]	2024

https://github.com/GuoJeff/generative-drug-design-with-experimental-validation

>70 studies with experimental validation (continuously updated)

Let’s switch general-purpose LLMs (à la GPT/Claude/Gemini)

From LLMs to LLM agents

User query

Answer

Generated
word by word.

Ability to take actions

1. Reflect

loop

2. Act

3. Observe

- Reduce hallucination
- Better tools, better task solving

ReAct: Synergizing Reasoning and Acting in Language Models
Yao et al., ICLR 2023

External tools

- Database query
- Web search
- Chemistry tools
- …

nature

https://doi.org/10.1038/s41586-026-10652-y

Accelerated Article Preview

A multi-agent system for automating scientific discovery

Received: 23 May 2025

Accepted: 12 May 2026

Accelerated Article Preview
Published online: 19 May 2026

Ali Essam Ghareeb, Benjamin Chang, Ludovico Mitchener, Angela Yiu, Caralyn J. Szostkiewicz, Dmytro Shved, Gavin J. Gyimesi, Jon M. Laurent, Samantha M. Wright, Muhammed T. Razzak, Andrew D. White, Silvia C. Finnemann, Michaela M. Hinks & Samuel G. Rodriques

nature

https://doi.org/10.1038/s41586-026-10644-y

Accelerated Article Preview

Accelerating scientific discovery with Co-Scientist

Received: 20 March 2025

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Published online: 19 May 2026

Cite this article as: Gottweis, J. et al. Accelerating scientific discovery with Co-Scientist. *Nature* https://doi.org/10.1038/s41586-026-10644-y (2026)

Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, Tao Tu, Petar Sirkovic, Artiom Myaskovsky, Grzegorz Glowaty, Felix Weissenberger, Alessio Orlandi, Dan Popovici, Anil Palepu, Keran Rong, Ryutaro Tanno, Khaled Saab, Fan Zhang, Jacob Blum, Andrew Carroll, Kavita Kulkarni, Nenad Tomašev, Dina Zverinski, Ivor Rendulic, Elahe Vedadi, Florian Hasler, Luka Rimanic, Marina Boia, Ivan Budiselic, Ben Feinstein, Mathias Bellaiche, Tom Sheffer, Jan Freyberg, Jeremy Ratcliff, Ottavia Bertolli, Katherine Chou, Avinatan Hassidim, Burak Gokturk, Amin Vahdat, Yuan Guan, Vikram Dhillon, Eeshit Dhaval Vaishnav, Byron Lee, Tiago R. D. Costa, José R. Penadés, Gary Peltz, Yossi Matias, James Manyika, Demis Hassabis, Yunhan Xu, Pushmeet Kohli, Annalisa Pawlosky, Alan Karthikesalingam & Vivek Natarajan

nature

https://doi.org/10.1038/s41586-026-10658-6

Accelerated Article Preview

An AI system to help scientists write expert-level empirical software

Received: 13 September 2025

Accepted: 13 May 2026

Accelerated Article Preview
Published online: 19 May 2026

Cite this article as: Aygün, E. et al. An AI system to help scientists write expert-level empirical software. *Nature* https://doi.org/10.1038/s41586-026-10658-6 (2026)

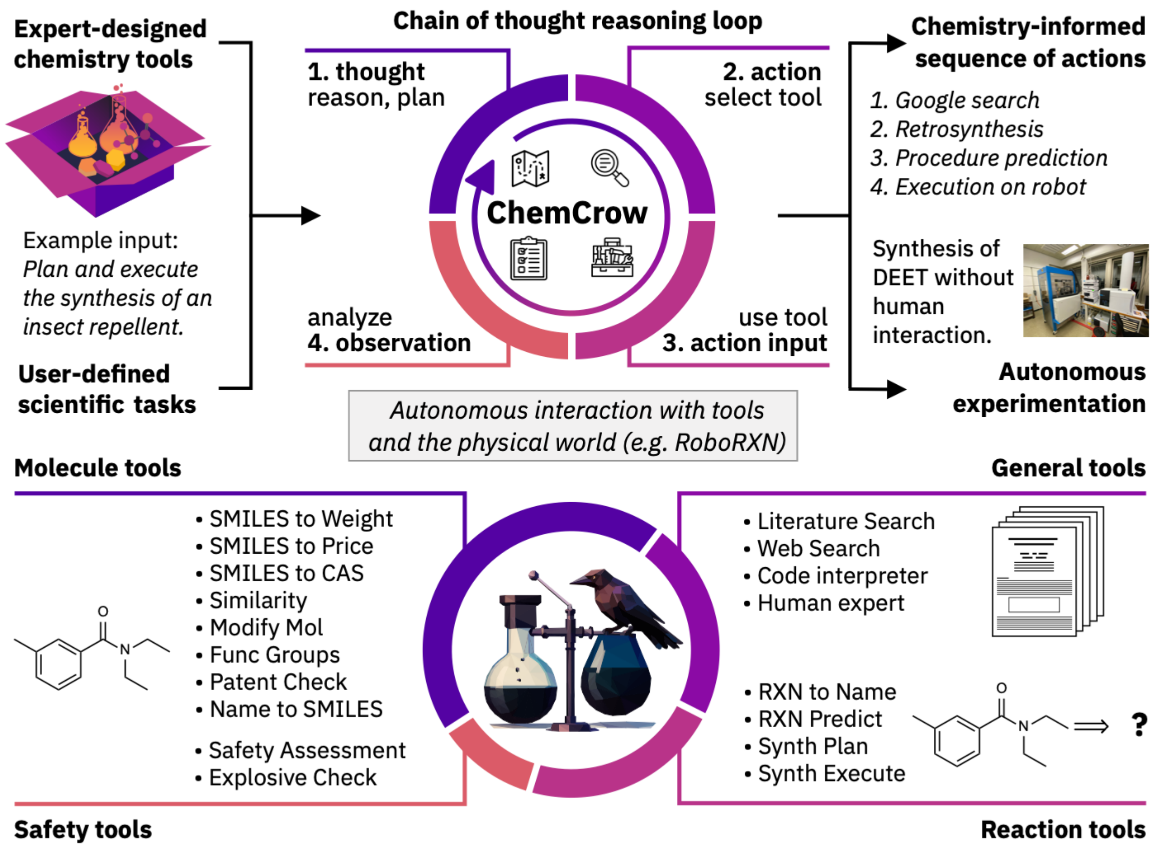
Eser Aygün, Anastasiya Belyaeva, Gheorghe Comanici, Marc Coram, Hao Cui, Jake Garrison, Renee Johnston, Anton Kast, Cory Y. McLean, Peter Norgaard, Zahra Shamsi, David Smalling, James Thompson, Subhashini Venugopalan, Brian P. Williams, Chujun He, Sarah Martinson, Martyna Plomecka, Lai Wei, Yuchen Zhou, Qian-Ze Zhu, Matthew Abraham, Erica Brand, Anna Bulanova, Jeffrey A. Cardille, Chris Co, Scott Ellsworth, Grace Joseph, Malcolm Kane, Ryan Krueger, Johan Kartiwa, Dan Liebling, Jan-Matthis Lueckmann, Paul Raccuglia, Xuefei Julie Wang, Katherine Chou, James Manyika, Yossi Matias, John C. Platt, Lizzie Dorfman, Shibl Mourad & Michael P. Brenner

Big month for LLM agents

- FutureHouse
- DeepMind (2x)

→ Not fully open, most likely by now you could get the same with 2-3 prompts on Codex/Claude

ChemCrow – Augmenting LLMs with Chemistry Tools



Collaboration with Andrew White (co-founded FutureHouse based on ChemCrow)

Bran A., Cox S., Schilter O., Baldassari C., White A., & Schwaller P., *Nat Mach Intell* **6**, 525–535 (2024). >950 citations

Example: From Prompt to Synthesis

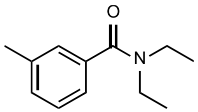
Task: Plan and execute the synthesis of an insect repellent

Thought: I need to know which insect repellent to synthesize. I will do a literature search to find a common insect repellent.
Action: LitSearch
Action input: common insect repellent

Observation: **DEET, picaridin, …**

Thought: I will choose DEET as the insect repellent to synthesize. First I need to get the SMILES of DEET.
Action: Name2SMILES
Action input: DEET

Observation: **CCN(CC)C(=O)c1cccc(C)c1**



Example: From Prompt to Synthesis

Thought: Now that I have the SMILES of DEET, I will plan the synthesis.
Action: RXNPlanner
Action input: **CCN(CC)C(=O)c1cccc(C)c1**
Observation:
To make CCN(CC)C(=O)c1cccc(C)c1, follow these steps:
1. **Add 0.3 mL of diethylamine to the reaction vessel.**
2. **Add 15 mL of dichloromethane to the reaction vessel.**
3. **Add 3.2 mmol of 3-methylbenzoyl chloride to the reaction vessel at 0°C.**
4. **Stir the mixture at 25°C for 1 hour […]**

RXN4Chemistry:
Recipe prediction

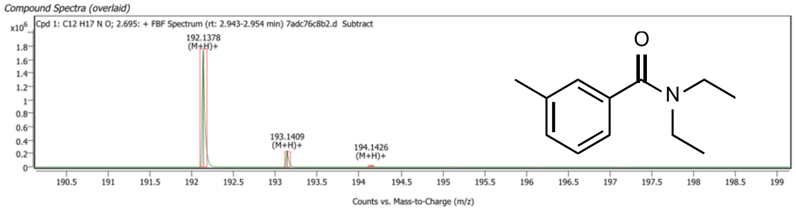
Thought: I have a synthesis plan for DEET. Now I will execute the synthesis.
Action: RXNExecute
Action input: **CCN(CC)C(=O)c1cccc(C)c1**

Observation: **Synthesis procedure completed!**

RoboRXN

Compound Summary

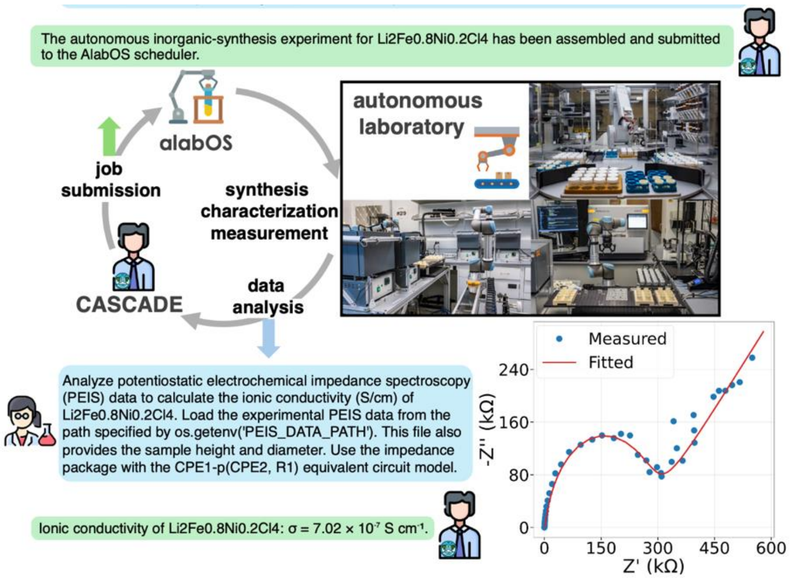
Cpd	Name	Formula	CAS	RT	Mass	Mass (Tgt)
1		C12 H17 N O		2.695	191.1305	191.1310
2		C4 H11 N		3.405	73.0896	73.0891



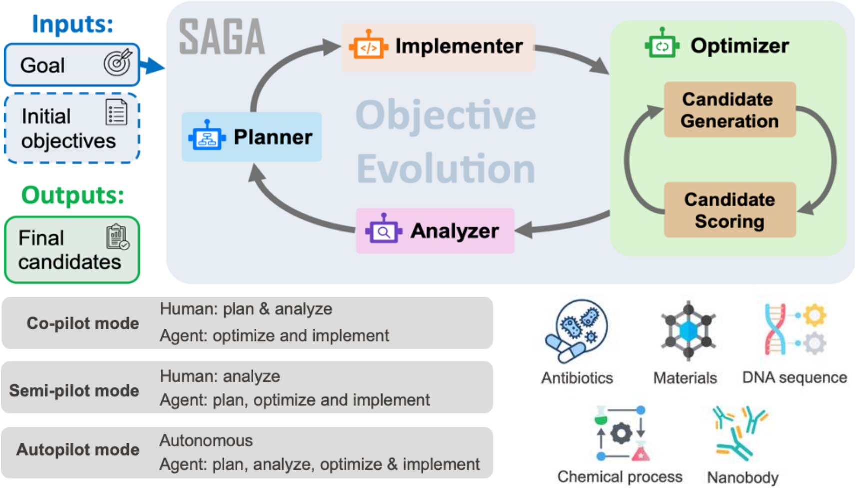
More recent agentic developments (collaborations)

CASCADE: Going from ”LLM + Tools” to **“LLM + Skill Acquisition”**

(Huang and Chen et al., with Ceder group and A-Lab, UC Berkeley, https://arxiv.org/abs/2512.23880)



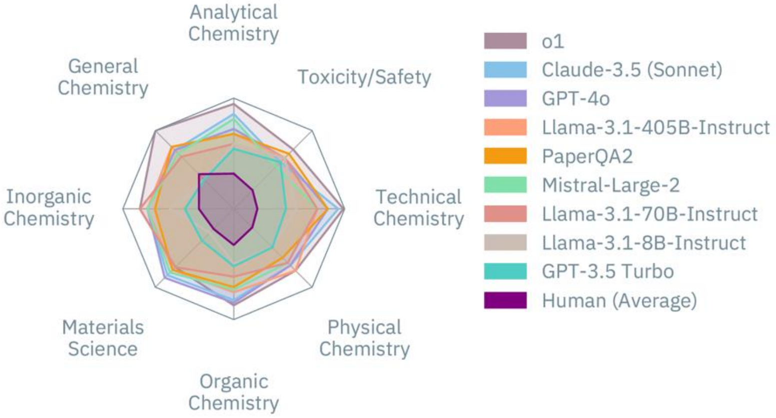
SAGA: Goal-evolving Agents
Large-scale collaboration with US groups (https://arxiv.org/abs/2512.21782)



DynaMate: Multi-Agent Framework for Protein-Ligand Molecular Dynamics Simulations. (Guilbert and Masschelein et al., https://arxiv.org/abs/2512.10034)

Evaluating LLMs/Agents is challenging

ChemBench

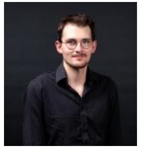


Article | Open access | Published: 20 May 2025

A framework for evaluating the chemical knowledge and reasoning abilities of large language models against the expertise of chemists

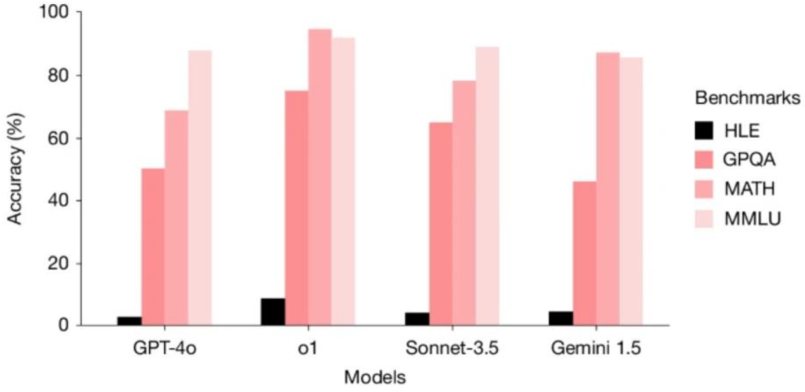
Nature Chemistry **17**, 1027–1034 (2025) | Cite this article

74k Accesses | **36** Citations | **91** Altmetric | Metrics



Collaboration led by Kevin Jablonka

Humanity's Last Exam (HLE)



Article | Open access | Published: 28 January 2026

A benchmark of expert-level academic questions to assess AI capabilities

Center for AI Safety, Scale AI & HLE Contributors Consortium

Nature **649**, 1139–1146 (2026) | Cite this article

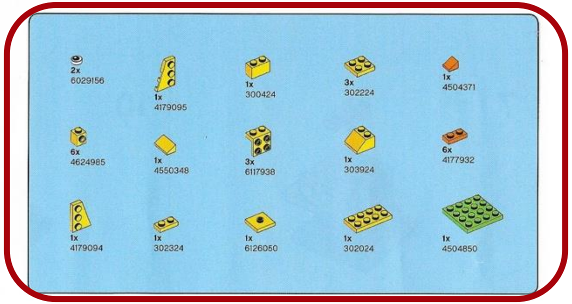
47k Accesses | **154** Altmetric | Metrics

Moving beyond multiple choice questions...

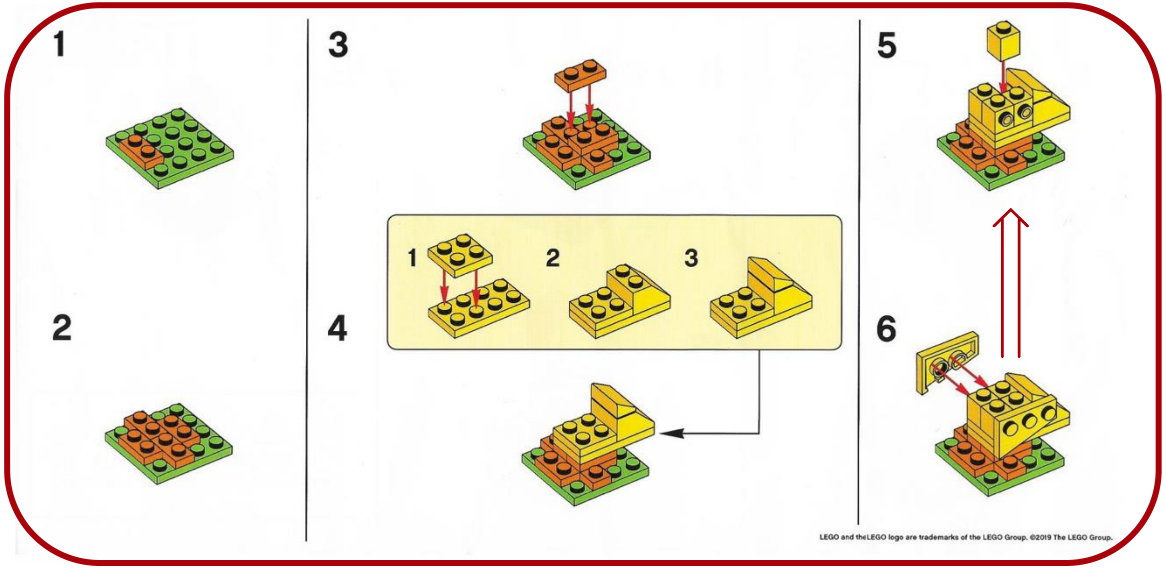
Retrosynthesis (Corey, Nobel prize, 1990)

Target molecule

Known (commercially available) building blocks

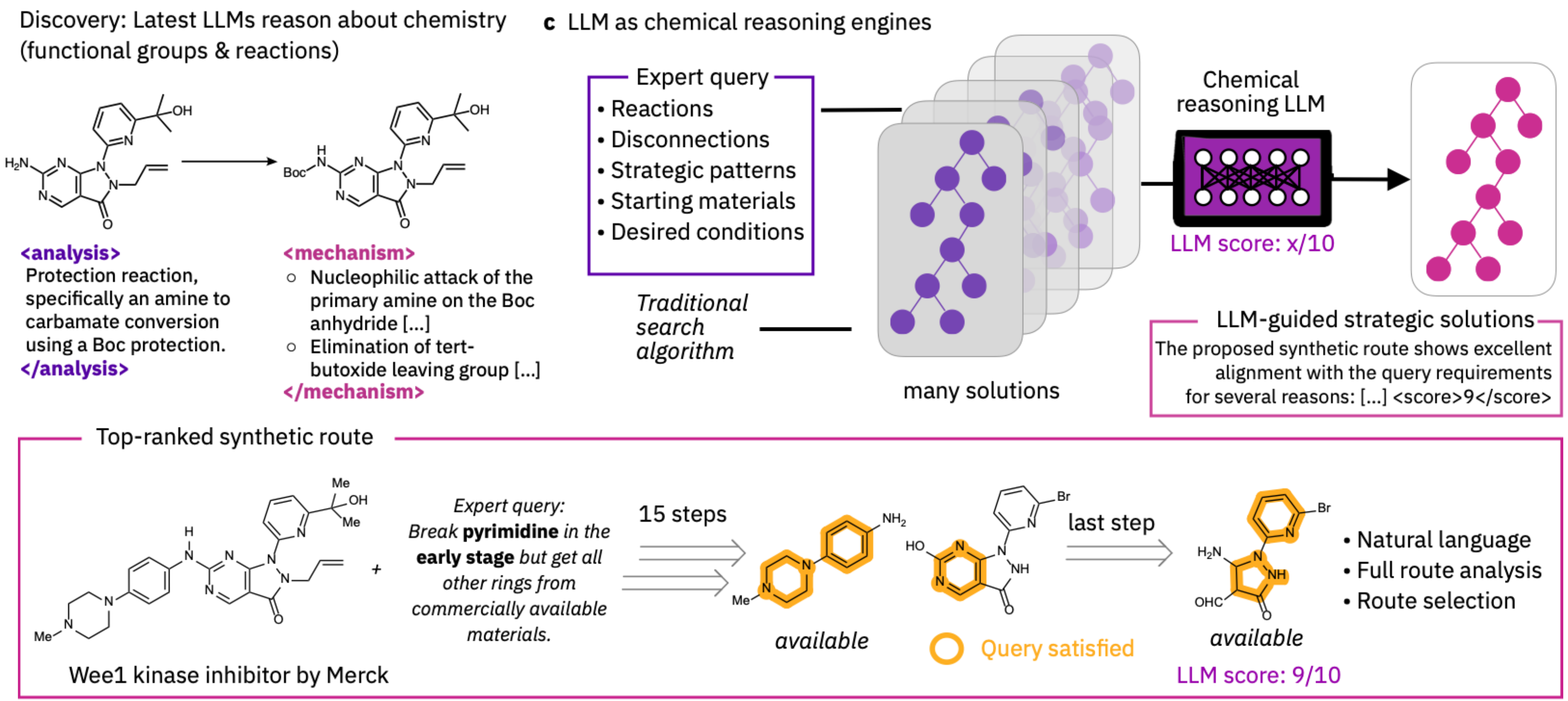


Steps to construct the target



Lego analogy: Amol Thakkar

LLMs as chemical reasoning engines

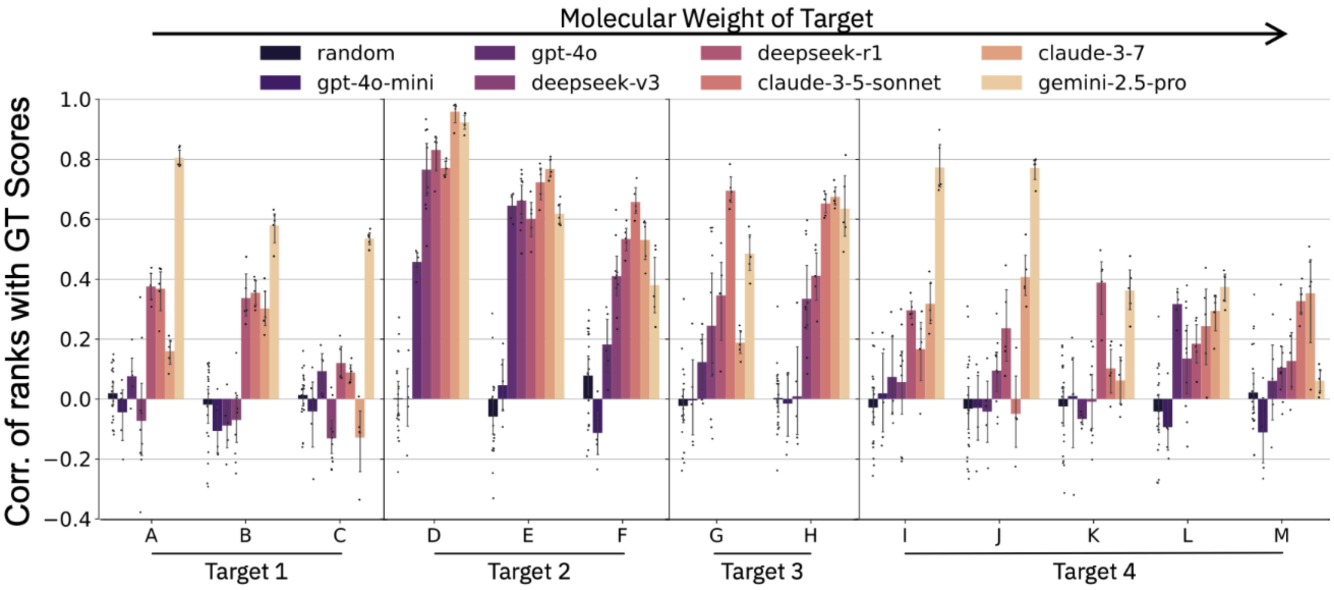


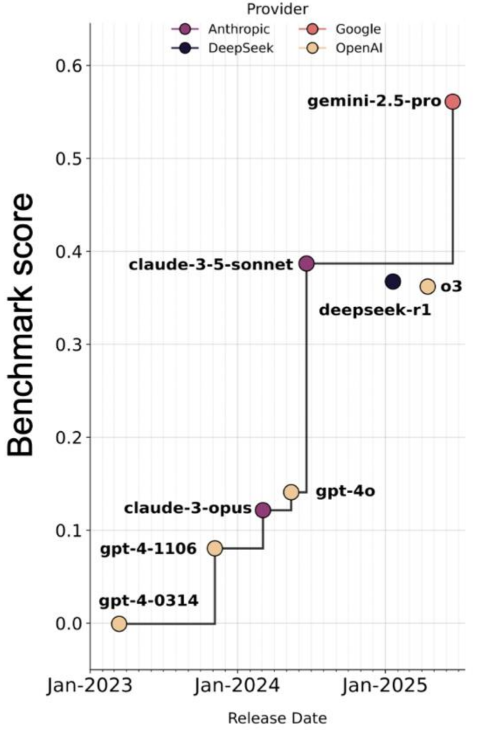
[Chemical reasoning in LLMs unlocks steerable synthesis planning and reaction mechanism elucidation]

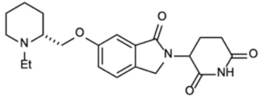
AM Bran, TA Neukomm, DP Armstrong, Z Jončev, P Schwaller

arXiv preprint arXiv:2503.08537

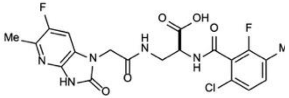
Strategy-aware synthesis planning benchmark



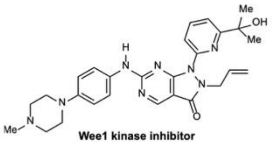




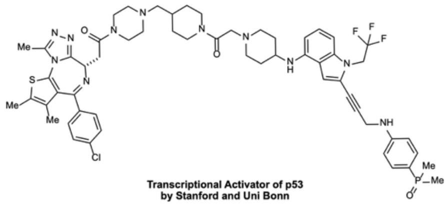
Molecular Glue Degrader
by Novartis



A Potent αvβ1 Integrin Antifibrotic
by Takeda



Wee1 kinase inhibitor
by Merck



Transcriptional Activator of p53
by Stanford and Uni Bonn

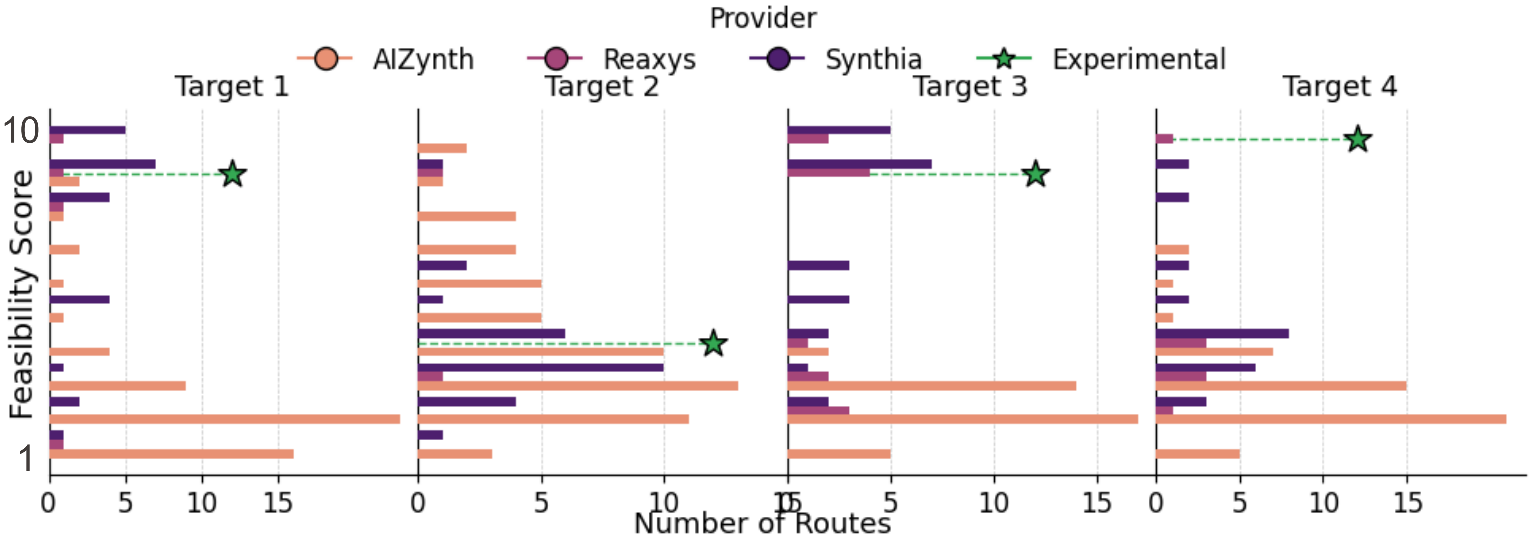
This approach enables chemists to talk to the synthesis planning tools in their language.

Feasibility assessment and ranking

Feasibility prompt

Highly feasible synthesis with high overall yields, consider potential side reactions and byproducts. Also ensure no unnecesary reactions are performed.

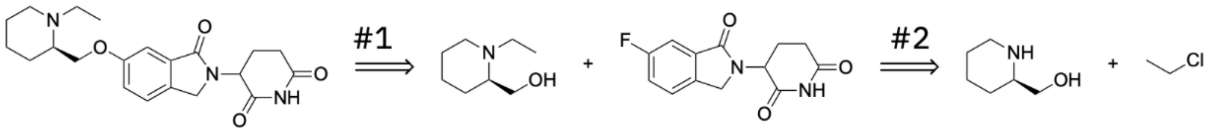
The strategy.



Plenty of predicted routes, only a few of high quality/feasibility.

Valuable and detailed feedback

Synthegy Score:
9/10



Reaction 1

The **SNAr reaction** on an activated aryl fluoride is a **robust and widely used** method in medicinal chemistry for C-O bond formation. Known to be highly feasible and generally proceeds in high yields (>80%).

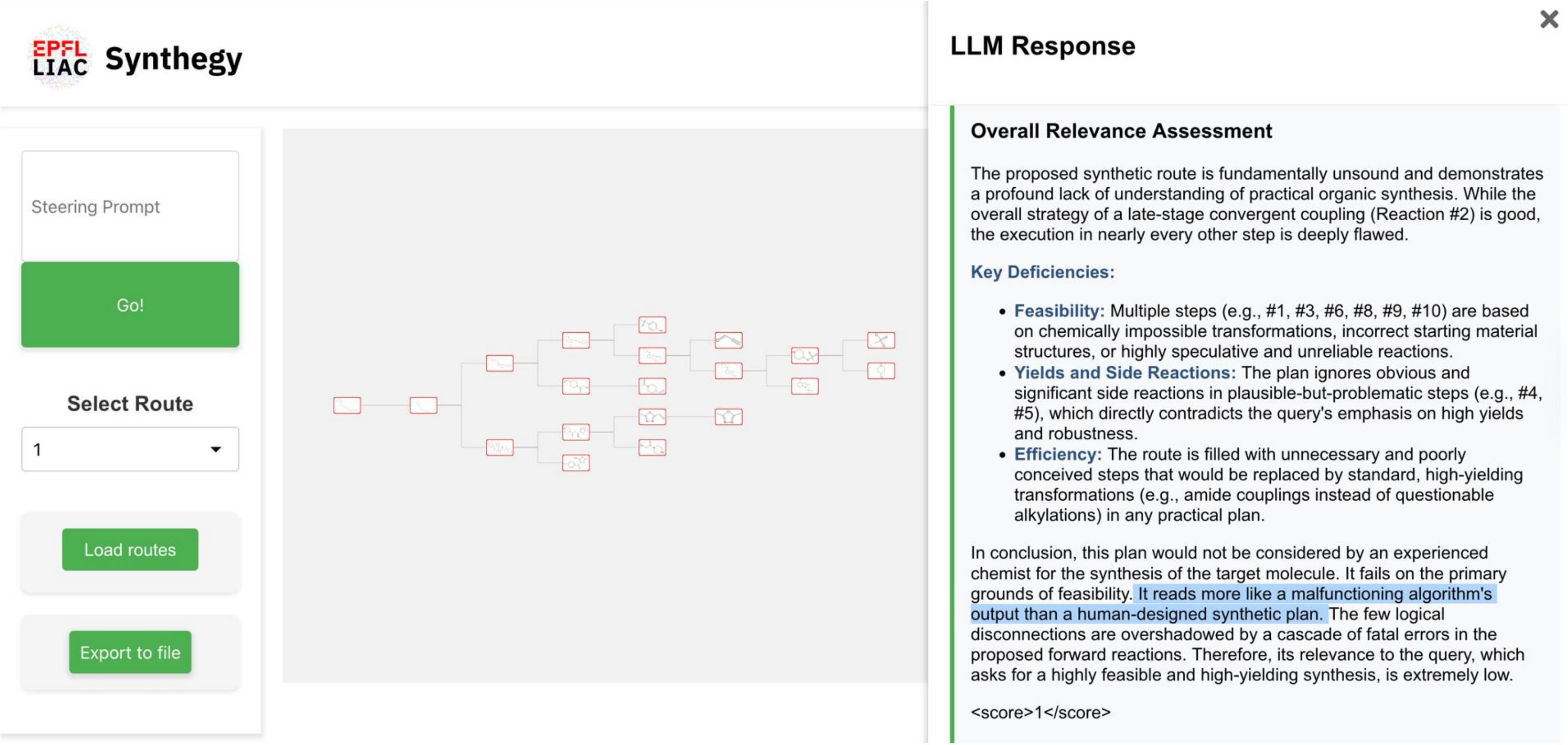
Reaction 2

Highly feasible, well-established. The primary alcohol on the reactant is a much weaker nucleophile than the secondary amine and is generally unreactive. **O-alkylation is a negligible side reaction.**

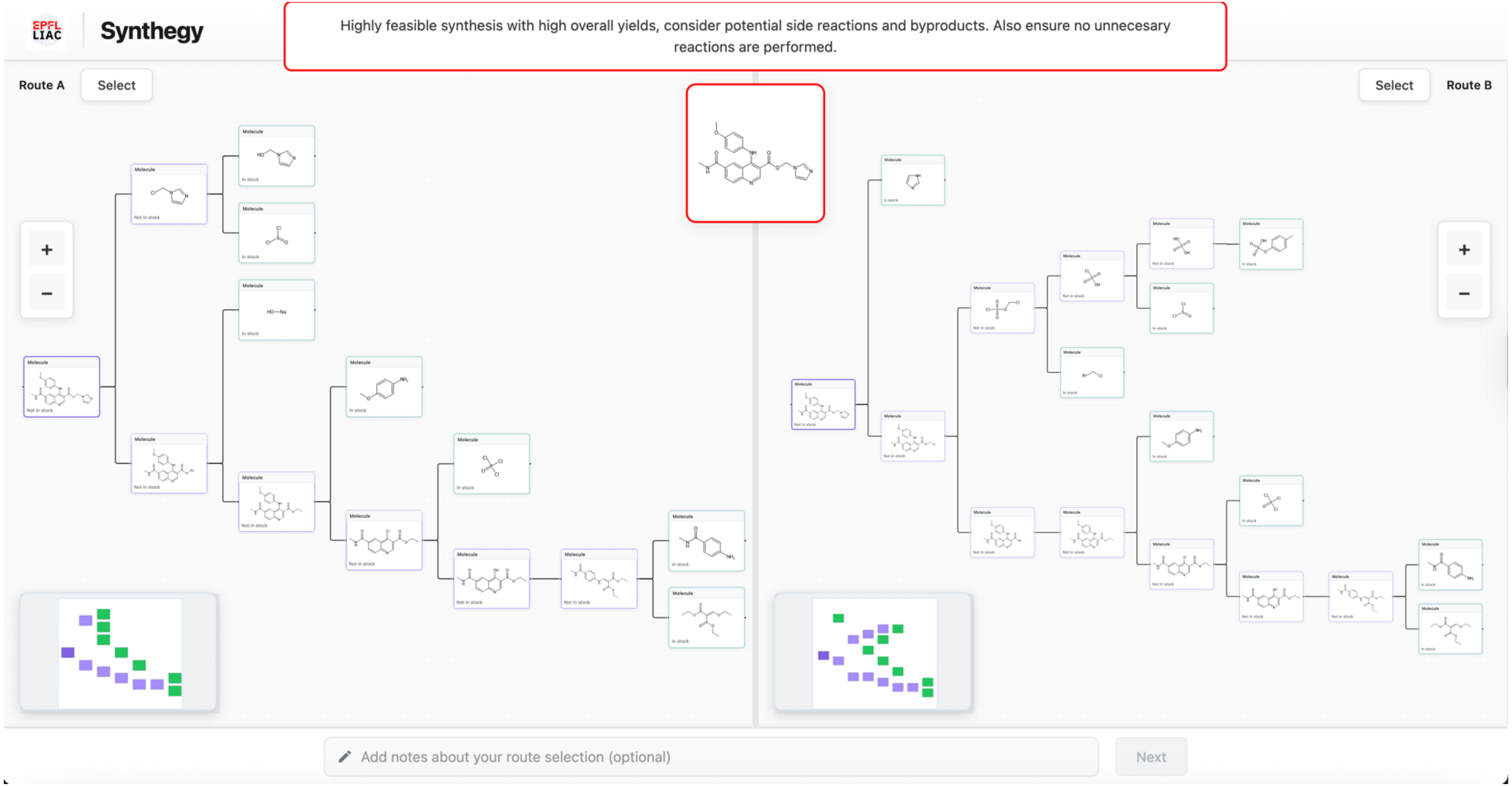
Overall Assessment

Modern, efficient, and robust approach to synthesizing the target. The **choice of a convergent strateg**y using powerful C-N and C-O bond-forming reactions is excellent. It directly addresses all aspects of the query: **high yields are expected**, side reactions are minimal and manageable, and the synthesis is concise.

LLMs can be harsh as synthetic chemists (valuable crap detectors)



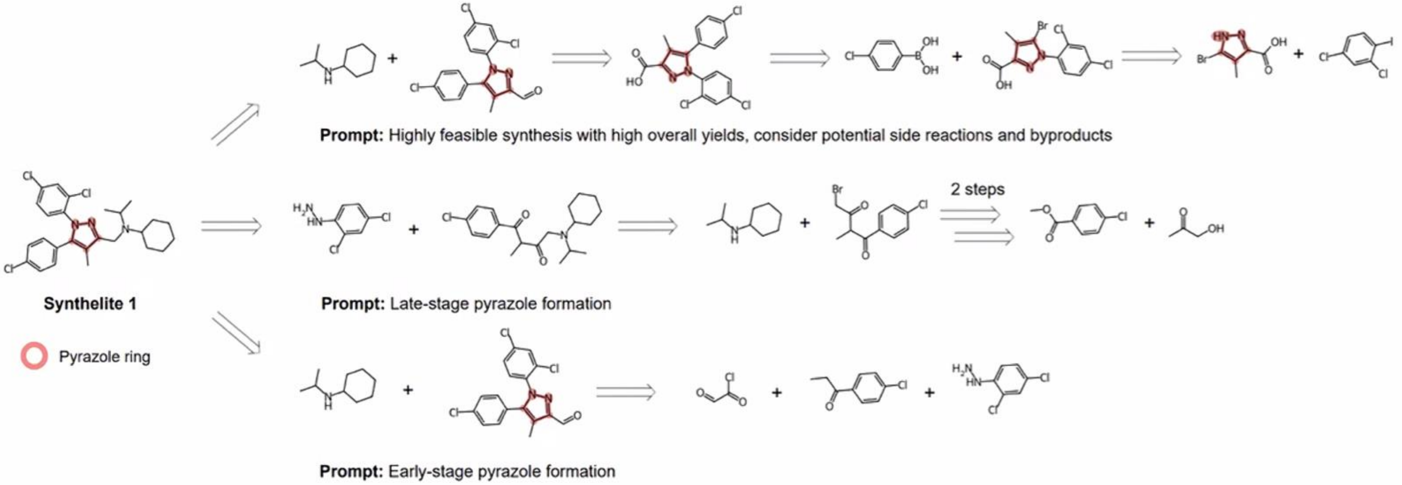
Time for a reality check – human expert evaluation



68 users (34 with >5 evaluations, 360 total) → Aligned: 259 (71.94%)
Our approach aligned with US star synthetic chemists (h-index >100) → 8/9

What’s next?

Synthelite – Agentic synthesis planning



Synthelite can map a short strategic prompt into full synthesis routes

- Incorporation of human expertise.
- Diverse exploration of solution space by prompting.

Synthelite: Chemist-aligned and feasibility-aware synthesis planning with LLMs

Nguyen Xuan-Vu, Daniel Armstrong, Milena Wehrbach, Andres M Bran, Zlatko Jončev, Philippe Schwaller

Now that you have a synthesis route, how to optimize single steps?

Article | Published: 03 February 2021

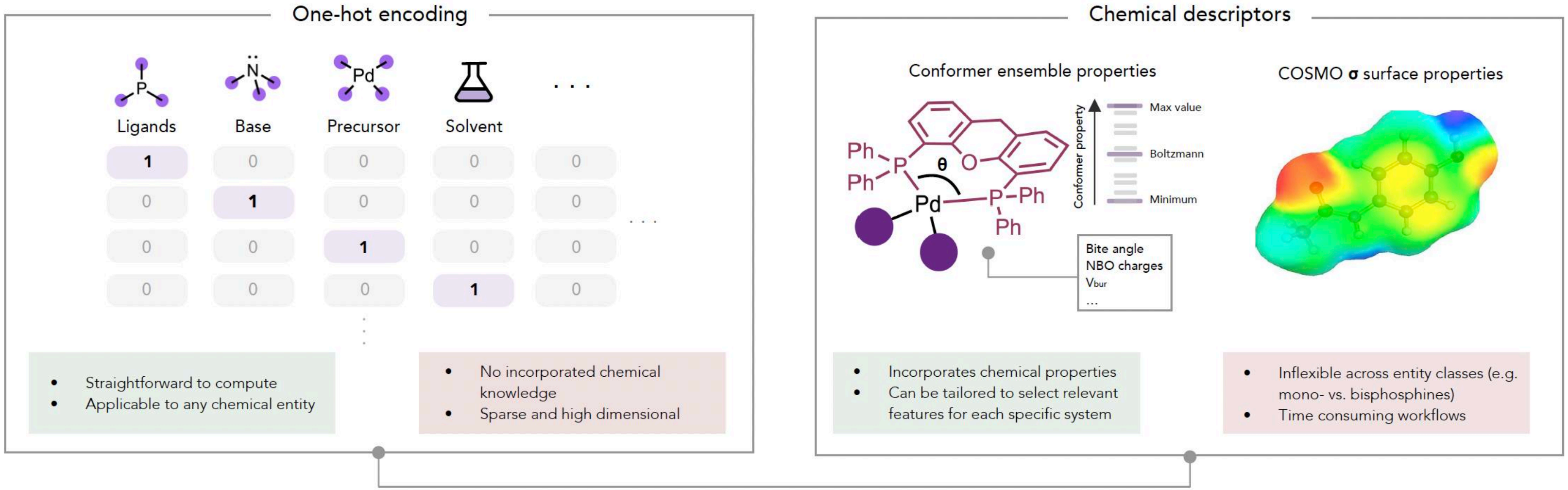
Bayesian reaction optimization as a tool for chemical synthesis

Benjamin J. Shields, Jason Stevens, Jun Li, Marvin Parasram, Farhan Damani, Jesus I. Martinez Alvarado, Jacob M. Janey, Ryan P. Adams & Abigail G. Doyle

Nature **590**, 89–96 (2021) | Cite this article

Representation: one-hot encoding or DFT descriptors

a Approaches for featurising chemical entities in model-driven reaction optimisation



Static representations which do not adapt to the optimisation task as more data is collected

Joshua Sin
(with Roche)

EPFL

LIAC's industrial collaborations

Joshua Sin
(with Roche)

Paulo Neves
(with Janssen)

Roche

nature communications

Article
https://doi.org/10.1038/s41467-025-61803-0

Highly parallel optimisation of chemical reactions through automation and machine intelligence

Received: 8 September 2024
Accepted: 25 June 2025

Joshua W. Sin[1,2], Siu Lun Chau[3], Ryan P. Burwood[4], Kurt Püntener[1], Raphael Bigler[1] & Philippe Schwaller[2,5]

Key point: **BO under HTE constraints** (large batches, temperature)
Representation: **Previous DFT descriptors (ligands)** + one hot
Data release: 1632 reactions data points (Ni-Suzuki/BH)

Roche

Swarm Intelligence for Chemical Reaction Optimisation

Rémi Schlama[1,2†], Joshua W. Sin[1,2*†], Ryan P. Burwood[3], Kurt Püntener[1], Raphael Bigler[1], Philippe Schwaller[2,4*]

Key point: **Swarm optimization as alternative to BO** for HTE
Representation: **Previous DFT descriptors (ligands)** + one hot
Data release: 989 reactions data points (Pd-Suzuki/Sulfonamide coupling) → Accepted in **Cell Chem**

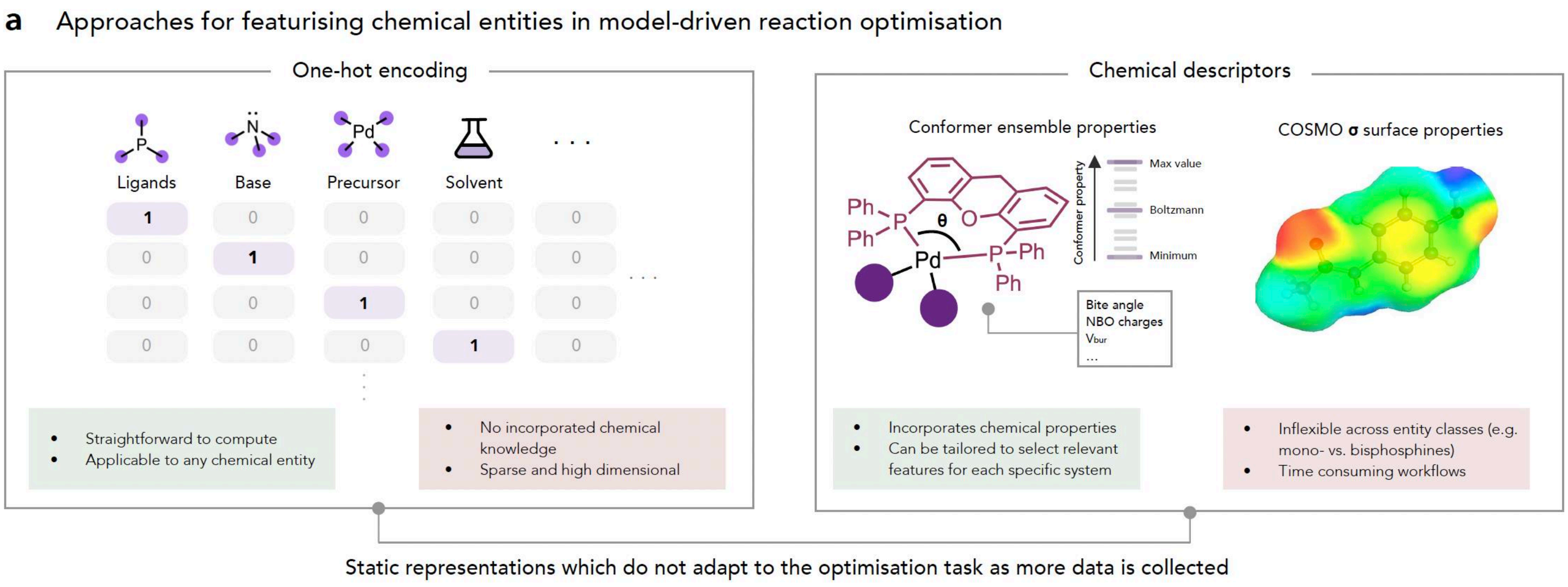
Janssen

Robust Out-of-Distribution Prediction of Buchwald-Hartwig Reactions

Paulo Neves[§1,2], Bo Hao[§3], Santeri Aikonen[§4], Justin B. Diccianni[3], Jörg K. Wegner[*6], Philippe Schwaller[*2,5], Iulia I. Strambeanu[*3]

Key point: **How to build datasets for predicting reaction succes**
Representation: **Previous DFT descriptors** / DRFP
Data release: 11328 reactions data points (BH)
→ Accepted in **Nature Comp Science**

For new reaction/process: What descriptors? DFT descriptors? Which ones? (you need data to validate good descriptors)



P. Schwaller (@SchwallerGroup)

Featurize reactions/procedures as text using LLMs

Jointly trained LLM (as Deep Kernel with PEFT) + GP

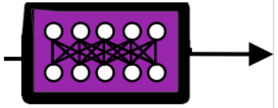
Reaction SMILES

```
CCc1ccc(I)cc1...COC(=O)
c1ccno1>>CCc1ccc(Nc2ccc
(C)cc2)cc1
```

Simplified procedure

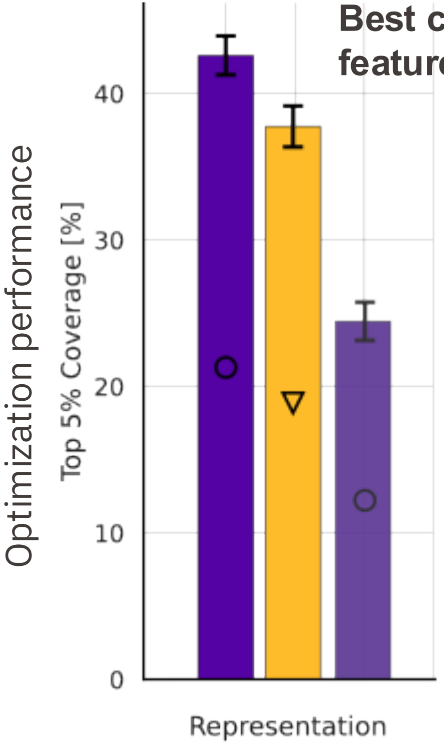
```
Reaction Setup
 - Reactant:
IC1=CC=C(CC)C=C1
 - Additive:
O=C(OC)C1=CC=NO1
        ...
```

LLM



[0.1, -0.03, …]

Everything can be encoded as TEXT.

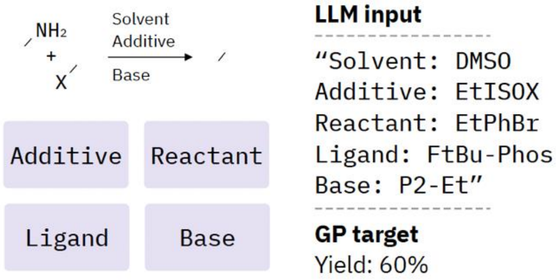


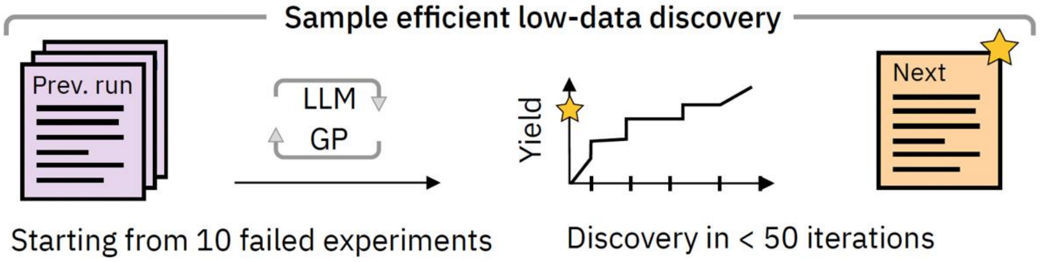
Large language models as uncertainty-calibrated optimizers for experimental discovery

- https://arxiv.org/abs/2504.06265

Bojana Ranković

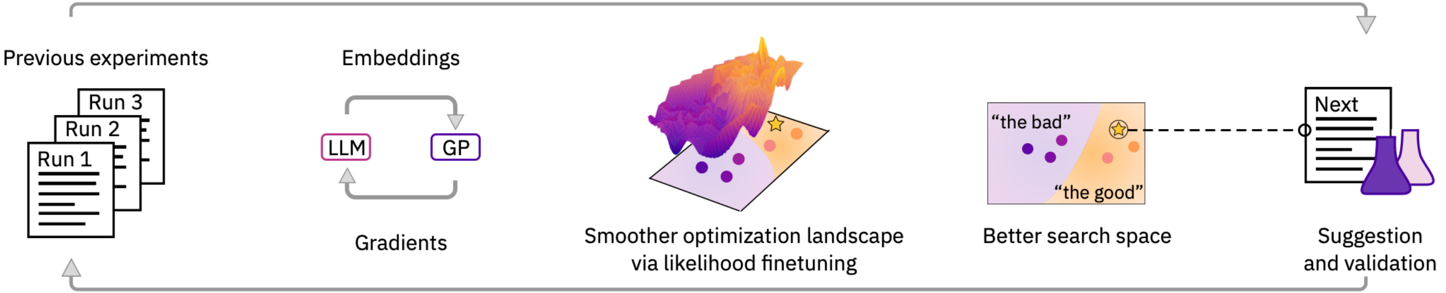
From language to likelihood



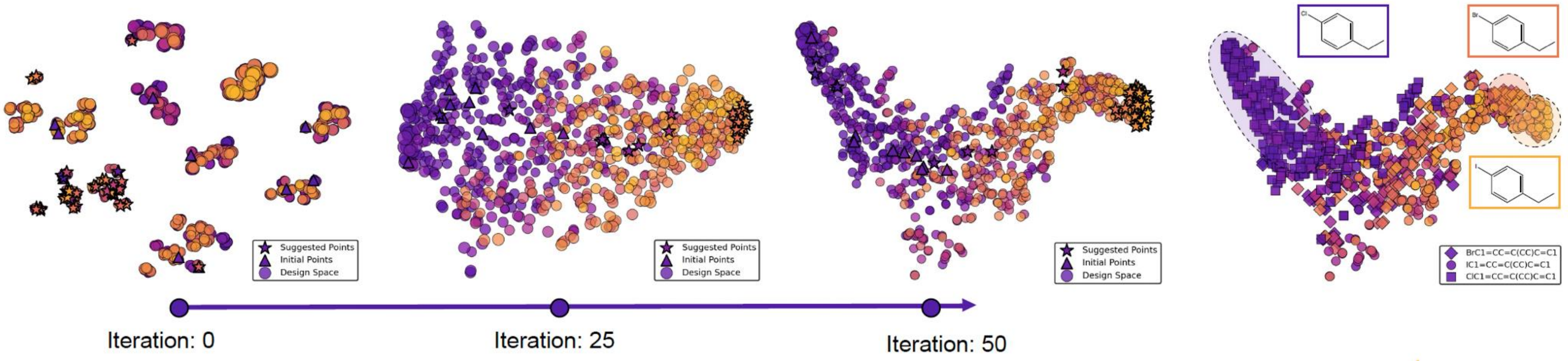


Experiments as text.

Initialization with failed experiments (below median)
→ **challenging task**

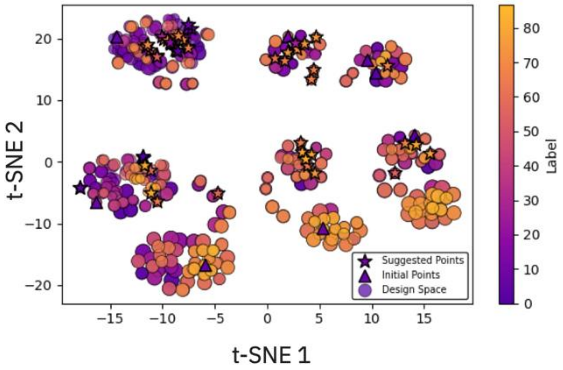


Low-data regime adaptation (reaction optimization)

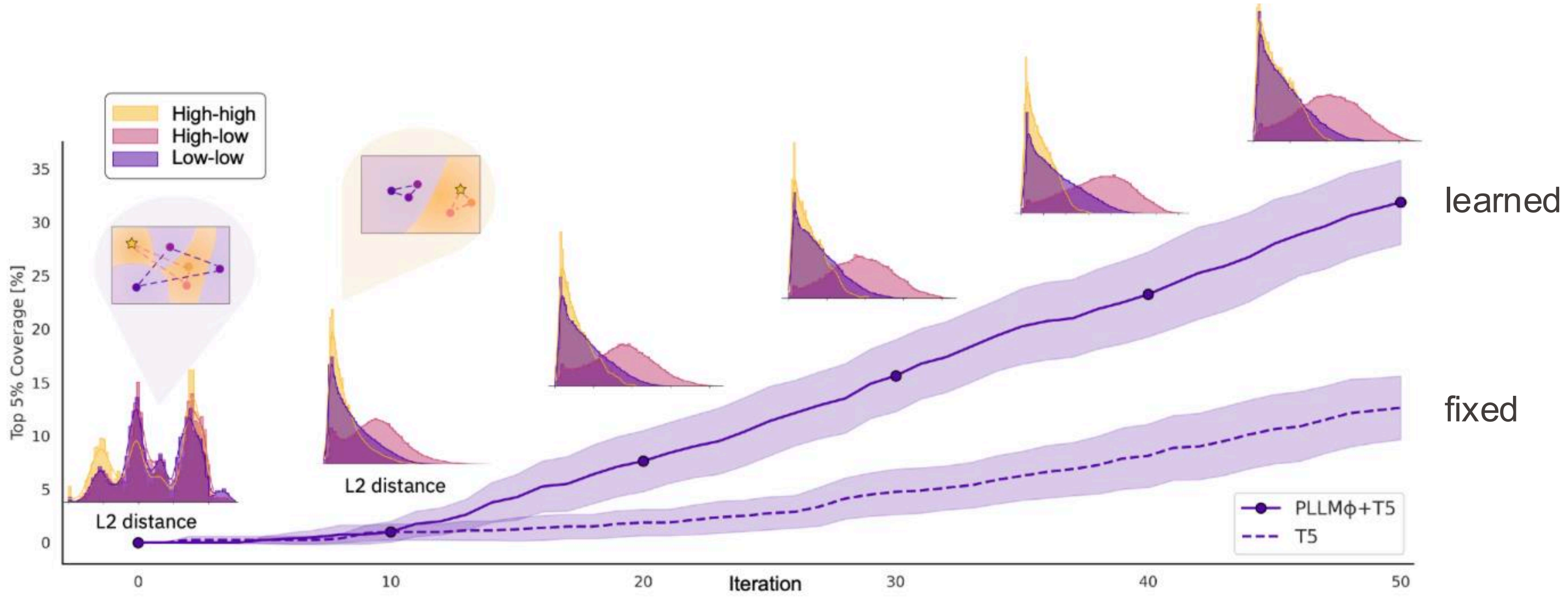


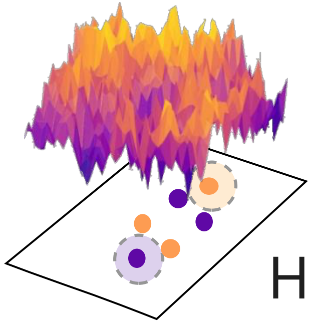
Adaptation to a new reaction space with a **few 10s of datapoints**

Static chemistry-informed representation:

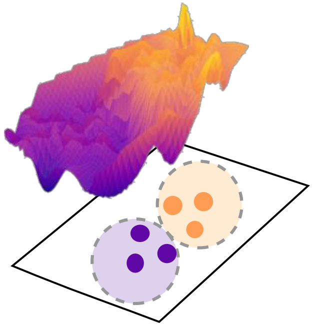


Learning a better representation → easier to optimize



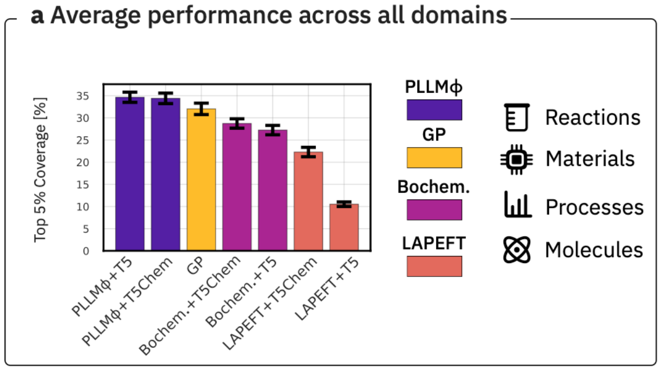


Hard to optimize



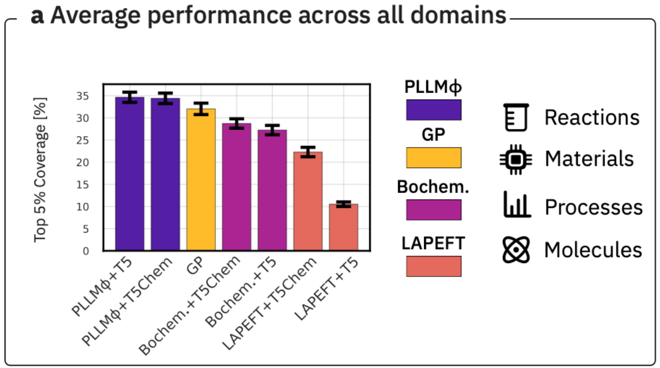
Easier to optimize

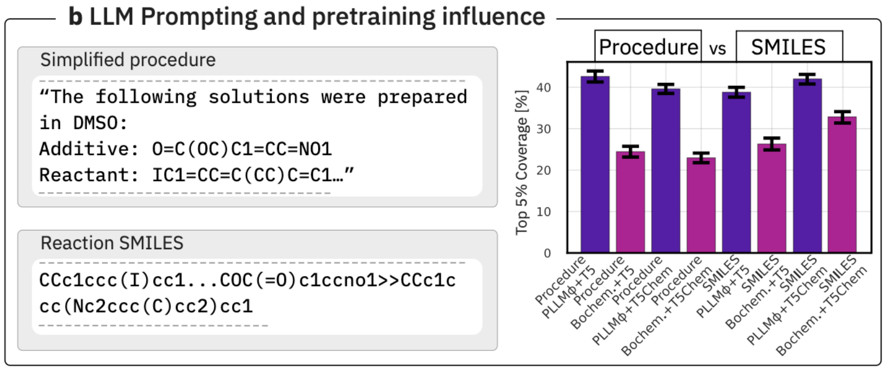
Cross-domain generality and performance



All results with a **single set of hyperparameters**
→ **No retuning required.**

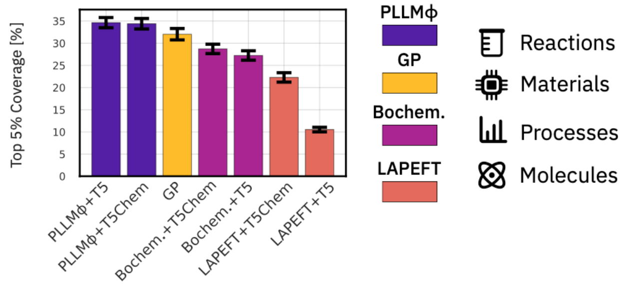
Cross-domain generality and performance





Cross-domain generality and performance

a Average performance across all domains



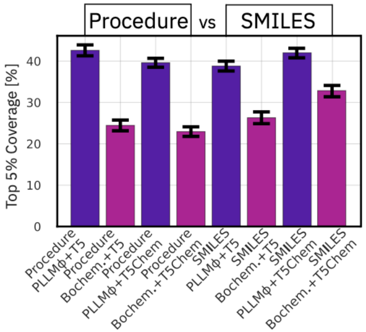
b LLM Prompting and pretraining influence

Simplified procedure

```
"The following solutions were prepared
in DMSO:
Additive: O=C(OC)C1=CC=NO1
Reactant: IC1=CC=C(CC)C=C1…"
```

Reaction SMILES

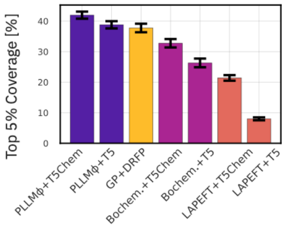
```
CCc1ccc(I)cc1...COC(=O)c1ccno1>>CCc1c
cc(Nc2ccc(C)cc2)cc1
```



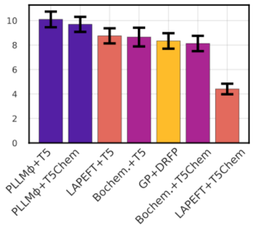
c Experimental optimization benchmarks across scientific domains

Synthetic organic chemistry

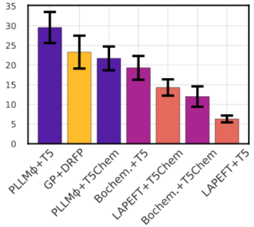
BH1-BH5



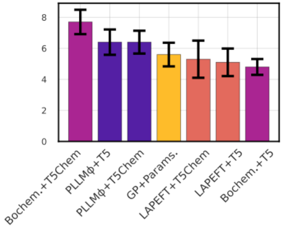
Additives 1-4



Suzuki Miyaura



Analytic and process chemistry

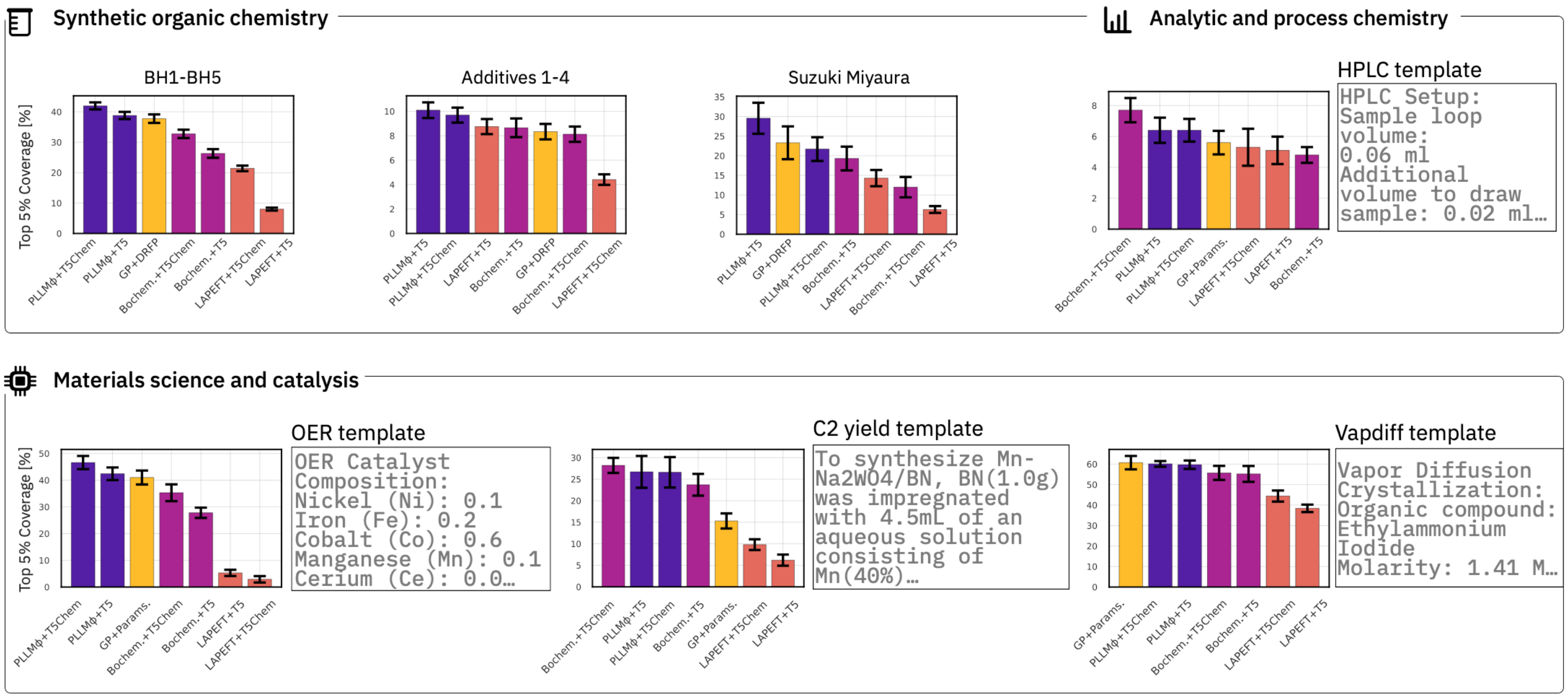


HPLC template

```
HPLC Setup:
Sample loop
volume:
0.06 ml
Additional
volume to draw
sample: 0.02 ml…
```

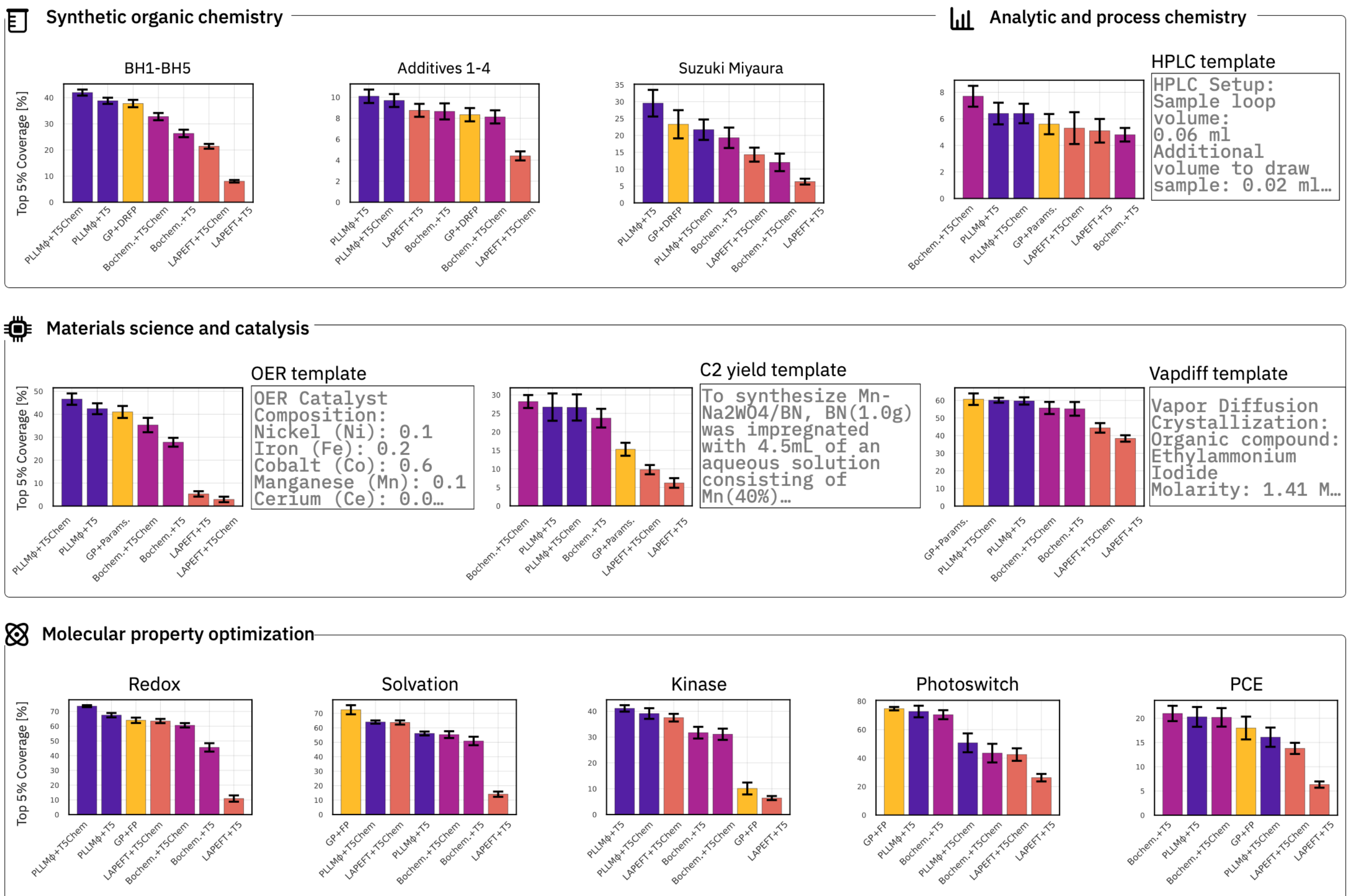
Cross-domain generality and performance

c Experimental optimization benchmarks across scientific domains



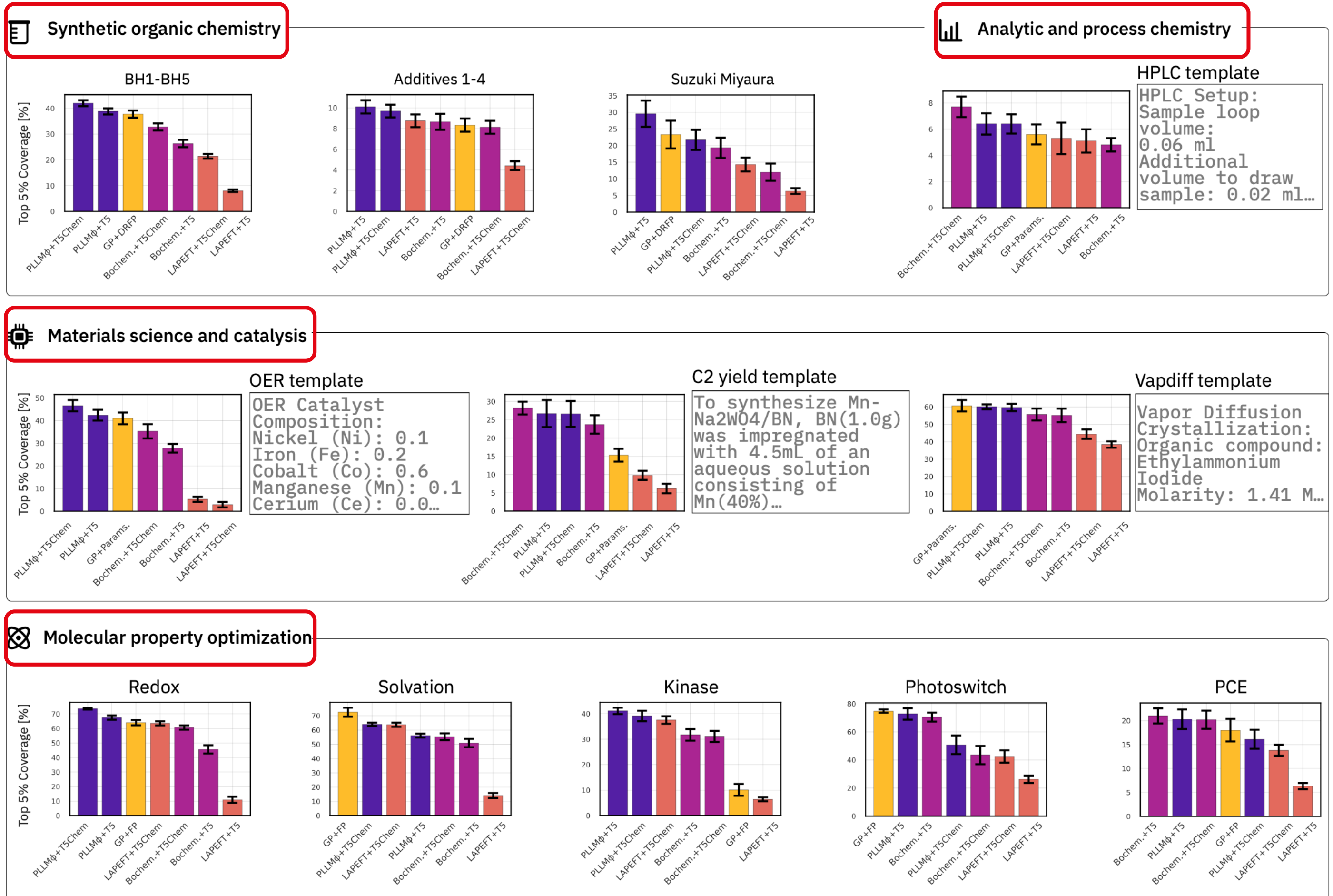
Cross-domain generality and performance

c Experimental optimization benchmarks across scientific domains



Cross-domain generality and performance

c Experimental optimization benchmarks across scientific domains



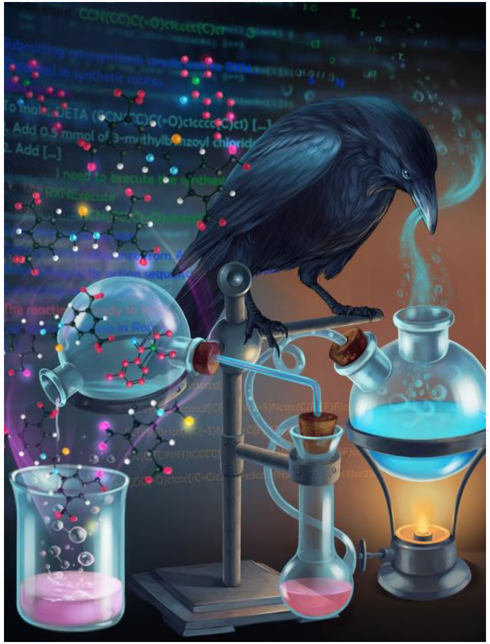
Everything can be represented as text...

→ Try GOLLuM on your optimization task.

https://github.com/schwallergroup/gollum

EPFL

Making Language Models Valuable in Chemistry and beyond



Small language models for sample-efficient molecular generation.

Large language models (LLMs) for chemical research and reasoning.

Low-data regime adaptation and representations.

Bridge from chemists' language to ML predictions

AI for Chem course from EPFL

- 01 - Basics
- 02 - Supervised Learning
- 03 - Intro to Deep Learning
- 04 - Unsupervised Learning
- 05 - Generative Models
- 06 - Generative Models 2
- 07 - Reaction Prediction
- 08 - Retrosynthesis
- 09 - Reaction properties
- 10 - Bayesian optimization
- 11 - Model deployment
- solutions

Contributors

This course is being created by the LIAC team. Many thanks to all the TAs:

- Cassandra Masschelein
- Remi Schlama
- Theo Neukomm

And all the old TAs participating to the creation of the notebooks:

- Victor Sabanza Gil
- Junwu Chen
- Bojana Ranković
- Andres CM Bran
- Jeff Guo

https://github.com/schwallergroup/ai4chem_course/tree/main/notebooks

LIAC *Team*

>15 nationalities — one team!
https://schwallergroup.github.io

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Many thanks for your attention!

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